

CS 305: Computer Networks

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Lecture 2: Introduction

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Chapter 1: roadmap

1.1 what is the Internet?

1.2 network edge

- end systems, access networks, links

1.3 network core

- packet switching, circuit switching, network structure

1.4 delay, loss, throughput in networks

1.5 protocol layers, service models

1.6 networks under attack: security

1.7 history

The network core

数据会被切分成一个个小的“包” (packet)，每个包独立传输，可能走不同的路径，最后在接收端重新组装

The mesh of **packet switches and links** that interconnects the Internet's end systems.

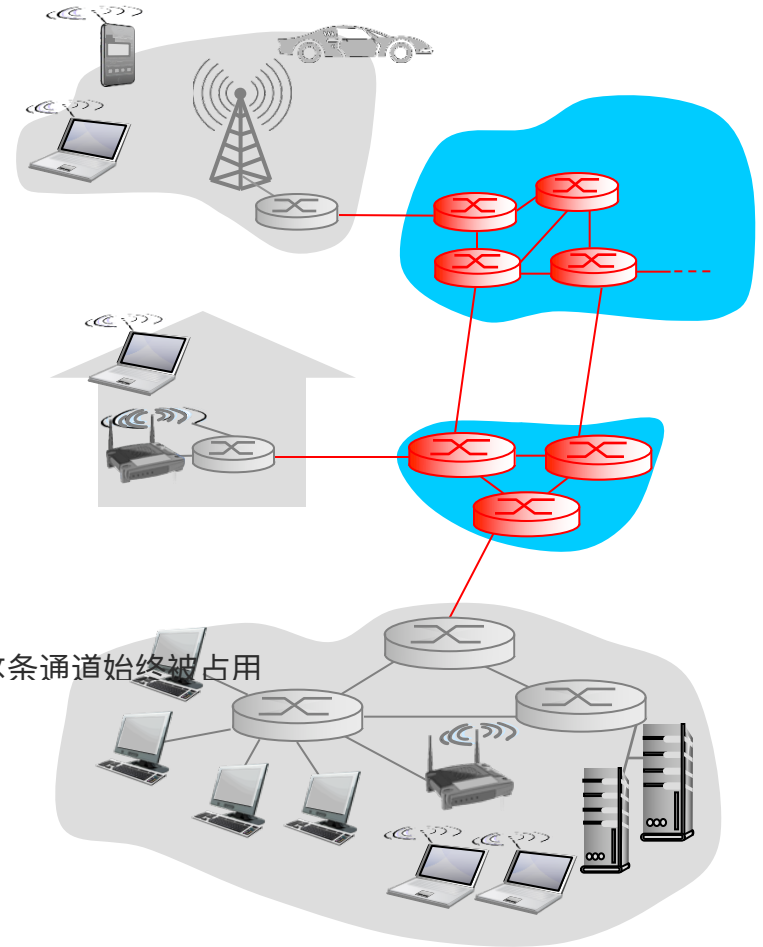
Packet switching:

- ❖ Internet
- ❖ not reserved (on demand); may wait

在通信前，先建立一条“电路”（类似于专属通道），通话过程中这条通道始终被占用

Circuit switching:

- ❖ telephone
- ❖ reserved



Packet-switching

Packet switching:

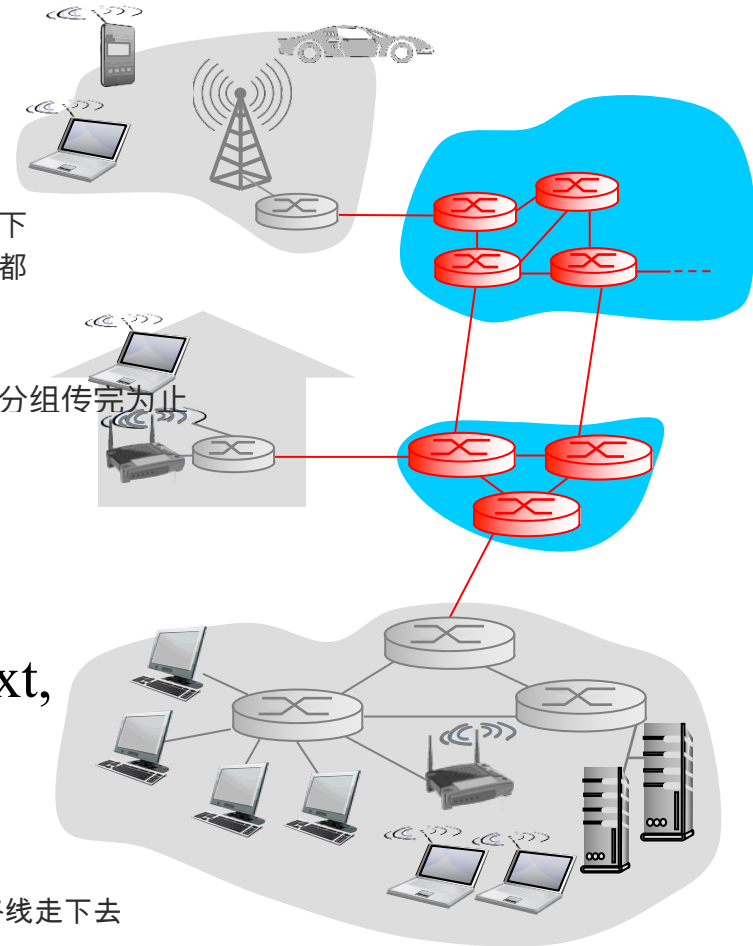
Hosts break long messages into packets;
each packet is forwarded independently

- **store-and-forward** 路由器接收到一个分组后，必须先完整存下来，然后再转发给下一个路由器；每一跳都要等一下
- each packet is transmitted **at full link capacity** 分组一旦上了链路，就会用到链路的全部带宽，直到分组传完为止
- **not reserved** → queueing delay and packet loss 如果队列满了，新的分组会被丢弃

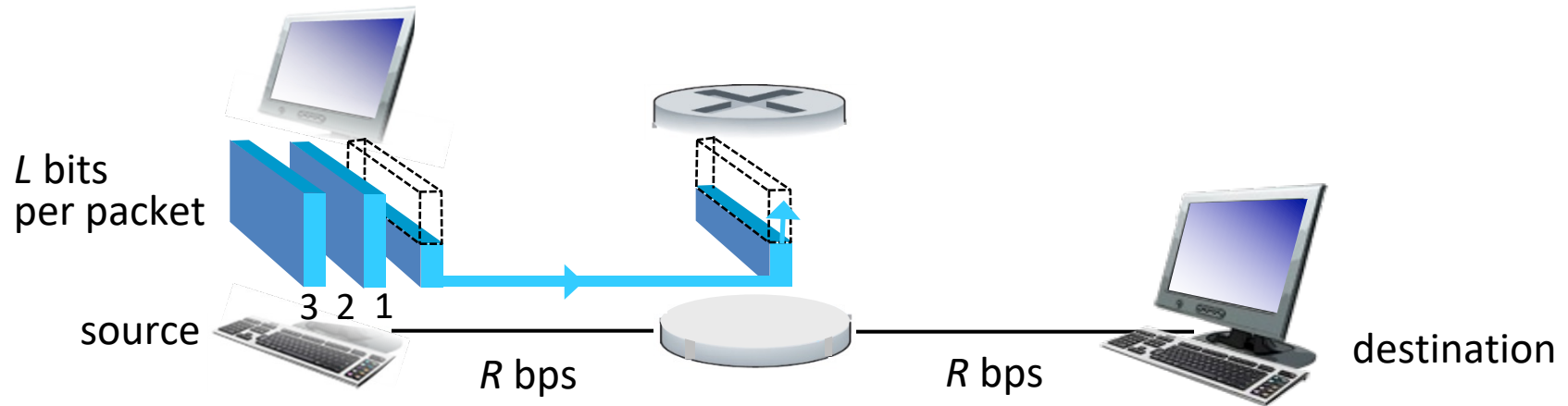
buffer缓存空间不足

Forward packets from one router to the next,
across links on path from source to
destination → **routing and forwarding**

routing : 规划路线 , forwarding : 一步一步按照路线走下去



Packet-switching: store-and-forward



- ❖ *Store and forward*: entire packet must arrive at router before it can be transmitted on next link
- ❖ Takes L/R seconds to transmit (push out) L -bit packet into link at R bps

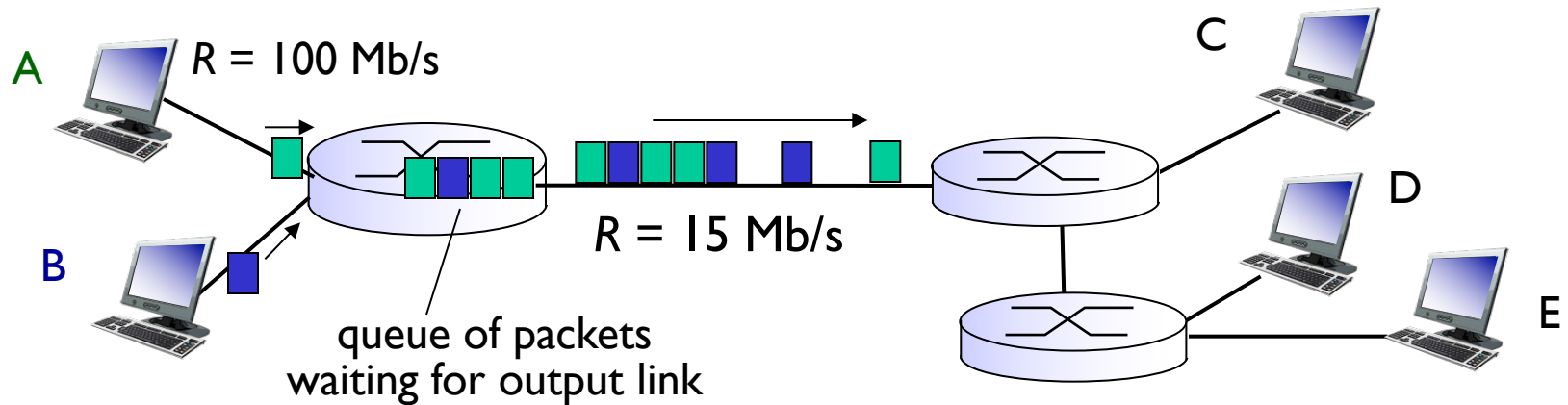
Example:

more on delay shortly ...

- $L = 7.5$ Mbits; $R = 1.5$ Mbps
- One-hop transmission delay = $L/R = 5$ sec
- End-end delay = $2L/R$ (assuming zero propagation delay)

How about one packet of length L sending over a path of N links, each of rate R ?

Packet switching: delay and packet loss



Output buffer stores packets that the router is about to send into that link.

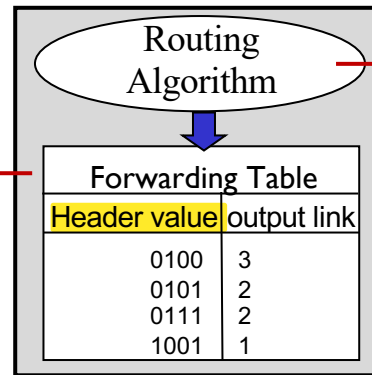
If **arrival rate (in bits) to link exceeds transmission rate of link** for a period of time:

- packets will queue, wait to be transmitted on link → **queuing delay**
- packets can be dropped (lost) if memory (buffer) fills up → **packet loss**

Packet switching: forwarding and routing

Forwarding:

local action: move arriving packets from router's input link to appropriate router output link



destination address in arriving packet's **header**

Routing:

- **global** action: determine source-destination paths taken by packets
- routing algorithms

分组 (packet) 的头部: 存放“地址信息”, 比如源地址、目的地址、端口号等

Circuit switching

Circuit switching: end-to-end resources **reserved** for “call” between source and destination (traditional telephone networks)

- ❖ Reserved (dedicated resources): buffer, link 电路一旦建立，带宽和缓冲区就锁定给这次会话。
- ❖ A fraction of each link's capacity 一条物理链路可以被划分成多个虚拟电路；不同用户分不同频率/时间片
- ❖ **Advantage:** **Guaranteed constant rate**
- ❖ **Limitation:** Circuit idle (空闲)
if not used by call (*no sharing*)
空闲不共享

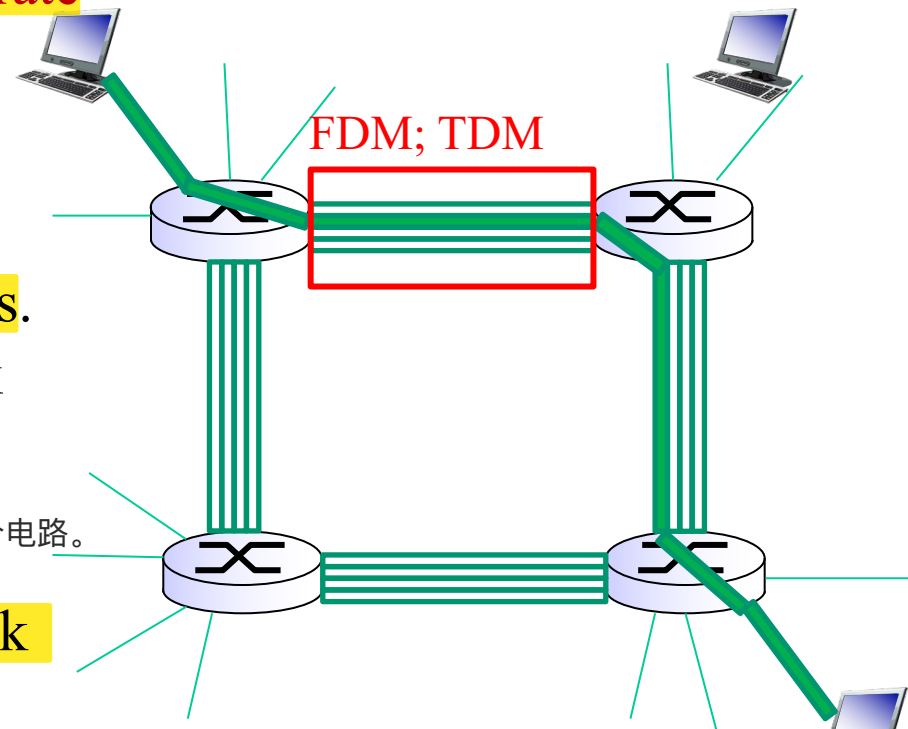
In diagram, **each link has four circuits.**

- The call gets 2nd circuit in top link and 1st circuit in right link.

某个通话可能在“上方链路”用第 2 个电路，在“右边链路”用第 1 个电路。

一旦分配好了 整个通话期间不会变化

How could we **split each physical link to multiple virtual links?**

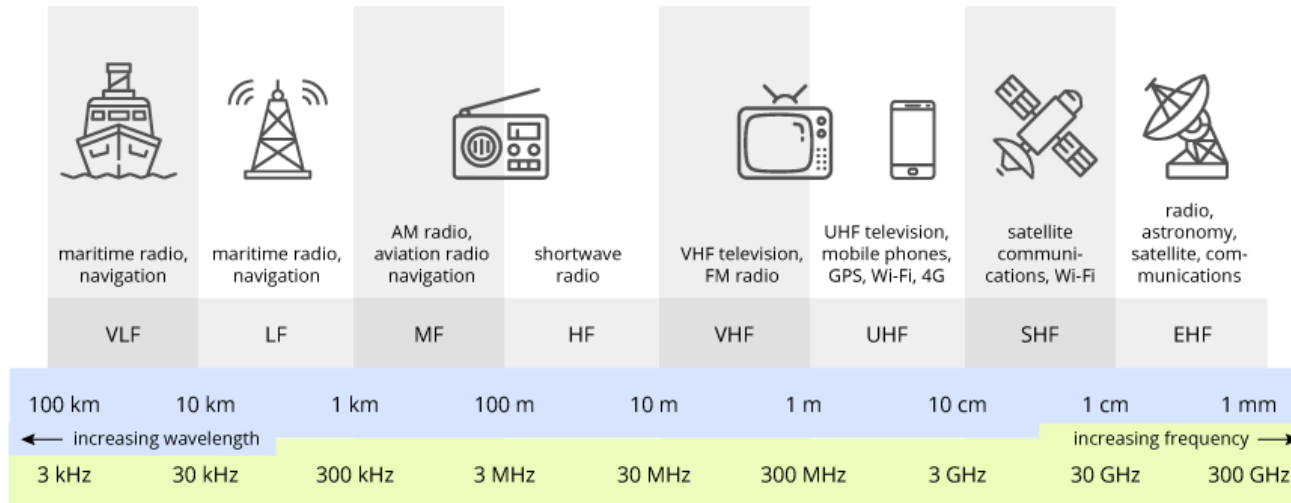
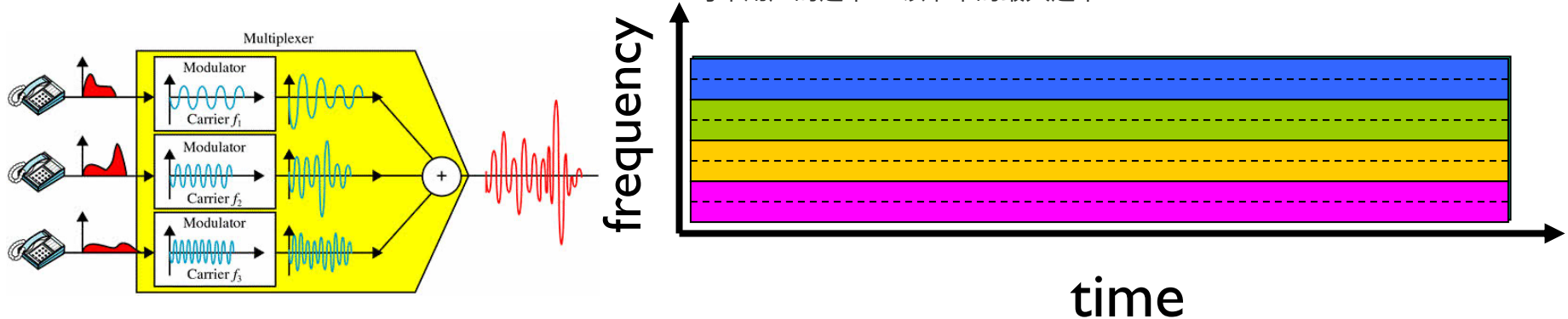
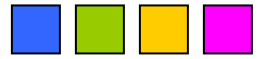


Circuit switching: FDM versus TDM

Frequency Division Multiplexing (多路复用) (FDM)

- Optical, electromagnetic frequencies divided into (narrow) frequency bands → **bandwidth**
- At max rate of that narrow band 每个用户的速率 = 该窄带的最大速率

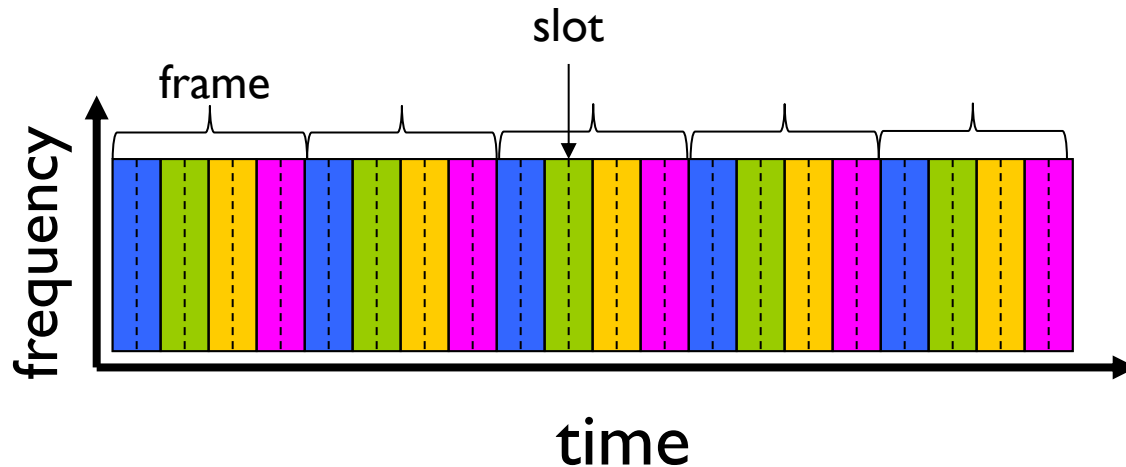
4 users



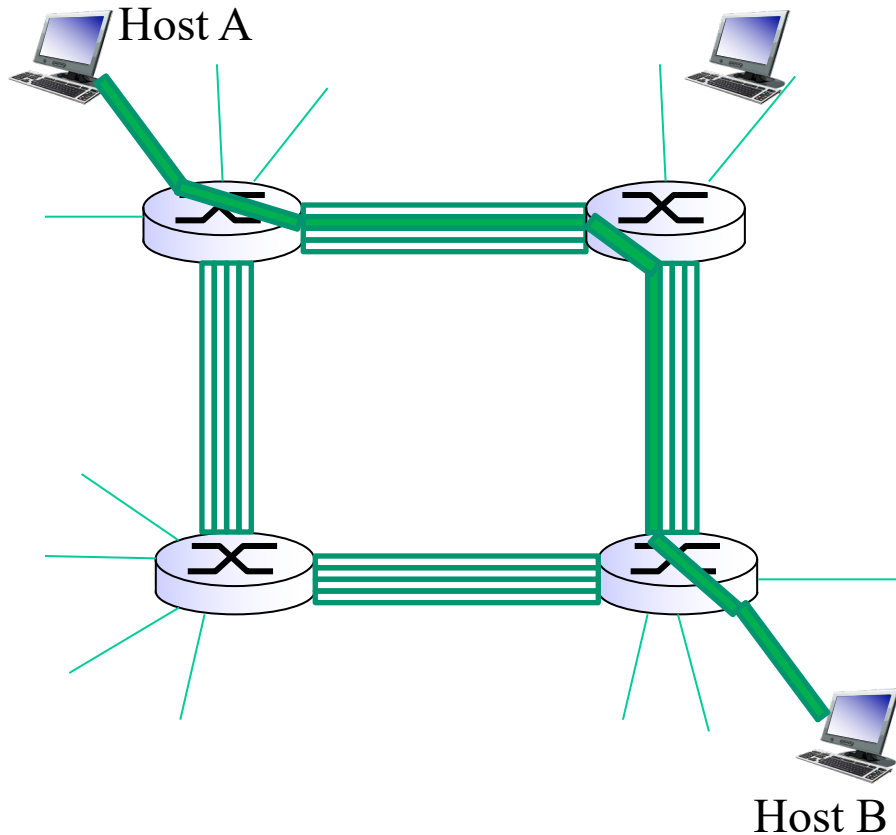
Circuit switching: FDM versus TDM

Time Division Multiplexing (TDM)

- Time divided into **slots** 每帧里分成多个时间片 (time slots). 不同用户轮流占用整个频带。
- At maximum rate of (wider) frequency band, but only during its time slot (s) 在“自己分到的时隙”里，用户可以用到整个带宽
- T slots in each frame: **T virtual links**



Circuit Switching: Delay



- A file of 640,000 bits
- All links in the network
 - use TDM with 24 slots per frame
 - a total bit rate of 1.536 Mbps
- It takes 500 msec to establish an end-to-end link.

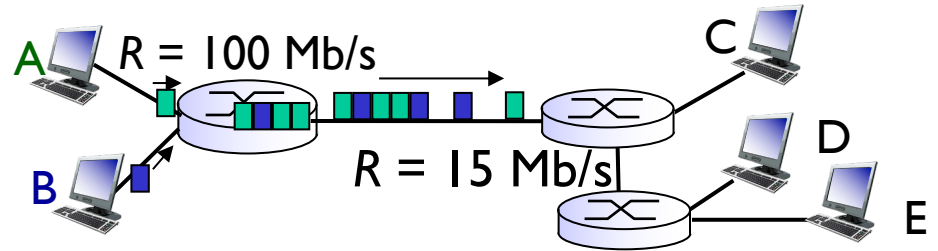
How long does it take to send the file?

- **Each circuit** has a transmission rate of $1.536 \text{ Mbps} / 24 = 64 \text{ kbps}$
- It takes $640,000 \text{ bits} / 64 \text{ kbps} = 10 \text{ sec}$
- Total: $10 \text{ sec} + 500 \text{ msec} = 10.5 \text{ sec}$

Packet Switching vs Circuit Switching

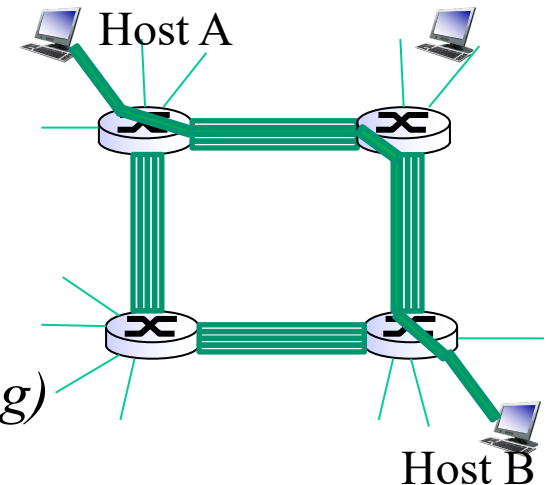
Packet switching:

- not reserved
- packet forwarded independently
 - store-and-forward
 - routing and forwarding
- at full link capacity
- packet loss, queuing delay



Circuit switching:

- reserved (dedicated resources): buffer, link, transmission rate
- establish an end-to-end link
- a fraction of each link's capacity
- guaranteed constant rate
- circuit segment idle if not used by call (*no sharing*)



Which is better?

Packet switching versus circuit switching

Packet switching allows more users to use network!

- ❖ *Users may be active with a small probability*
- ❖ **Circuit switching** pre-allocates use of the transmission link regardless of demand, with allocated but unneeded link time going unused.
- ❖ **Packet switching** on the other hand allocates link use *on demand*.

Example:

- 1 Mb/s link
- Each user:
 - At each time, active with a probability of 10%
 - 100 kbps when “active”

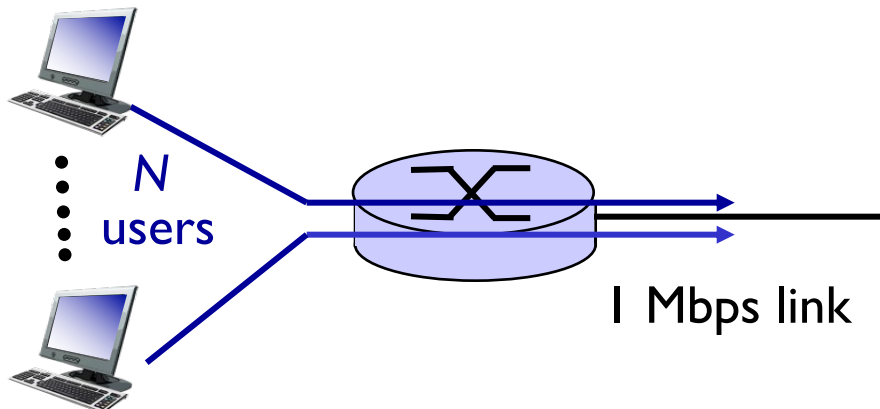
Circuit switching:

10 users

Packet switching:

With 35 users, that probability that more than 10 users are active at same time is less than 0.0004.

$$\sum_{n=11}^{35} 0.1^n \times 0.9^{35-n} \binom{35}{n}$$



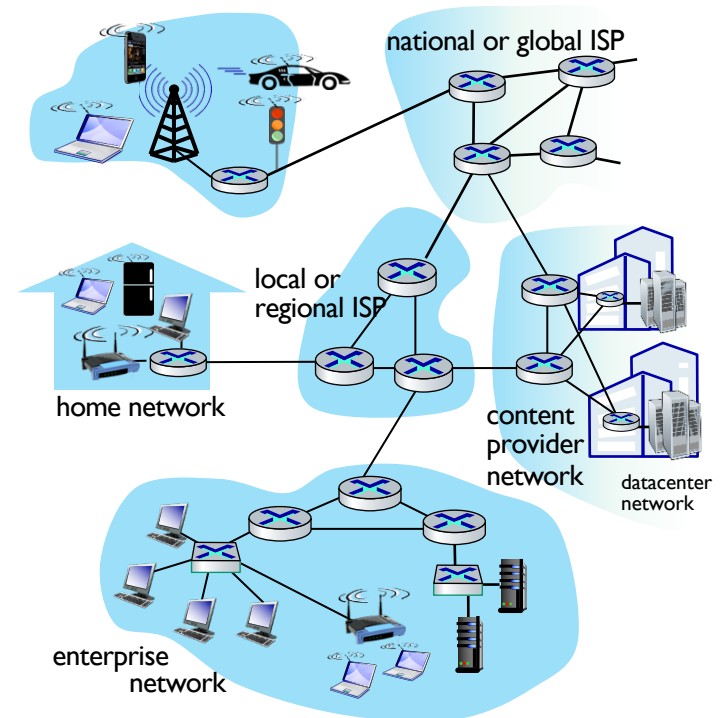
Packet switching versus circuit switching

Packet switching:

- ❖ great for bursty data 突发性数据
 - resource sharing
 - simpler, no call setup 不用建立占用路径
- ❖ **excessive congestion possible:** queuing delay and packet loss
 - protocols needed for reliable data transfer, congestion control
- ❖ **Q: How to provide circuit-like behavior?** 带宽保证
 - bandwidth guarantees needed for audio/video apps
 - still an unsolved problem (chapter 7)

Internet structure: network of networks

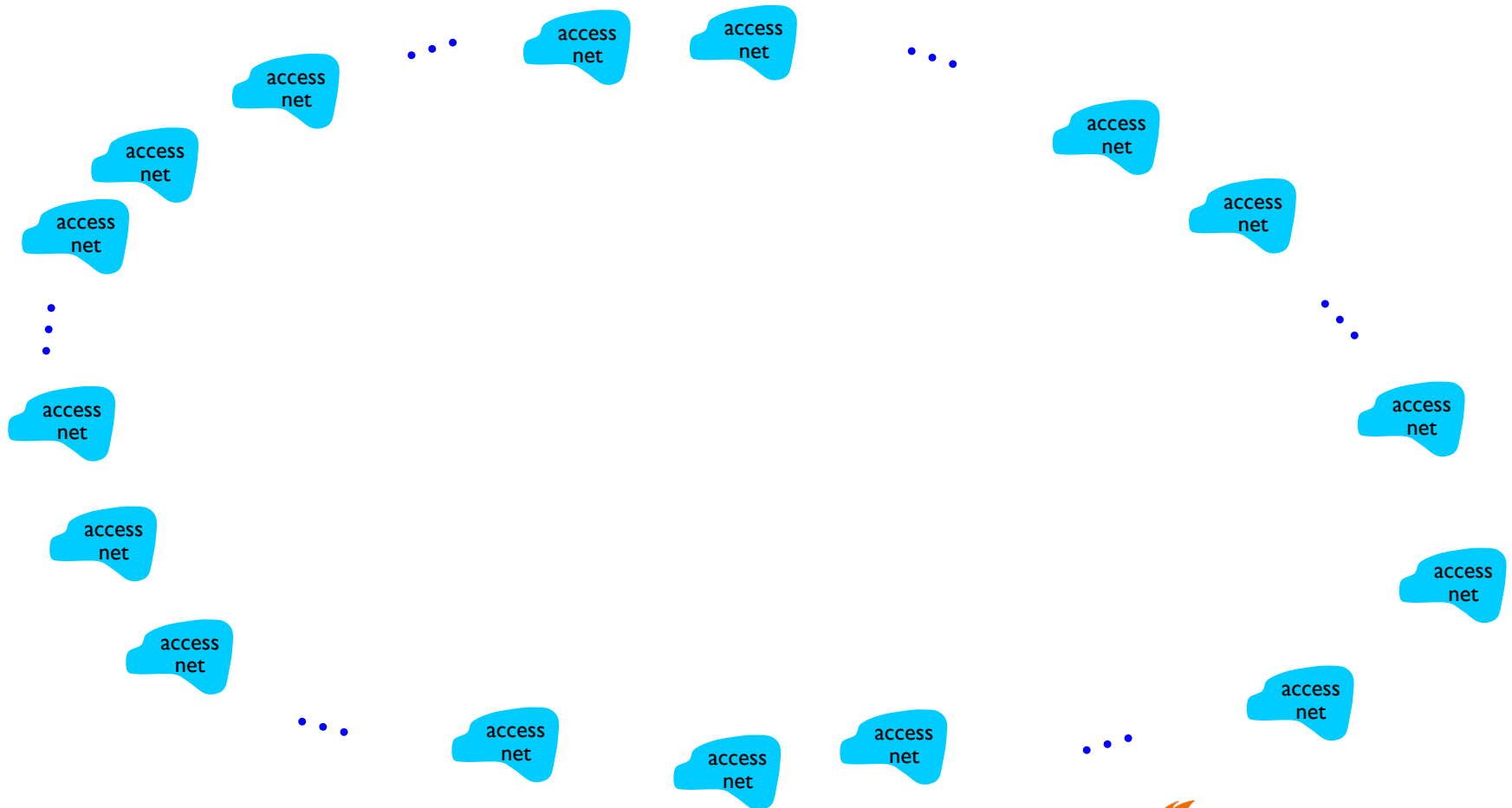
- ❖ End systems connect to Internet via **access ISPs** (Internet Service Providers)
 - Residential ISPs
 - University and company ISPs
- ❖ Access ISPs in turn must be **interconnected**.
 - ❖ So that any two hosts can send packets to each other
- ❖ Resulting network of networks is very complex
 - ❖ Evolution was driven by **economics** and **national policies**



Let's take a stepwise approach to describe current Internet structure.

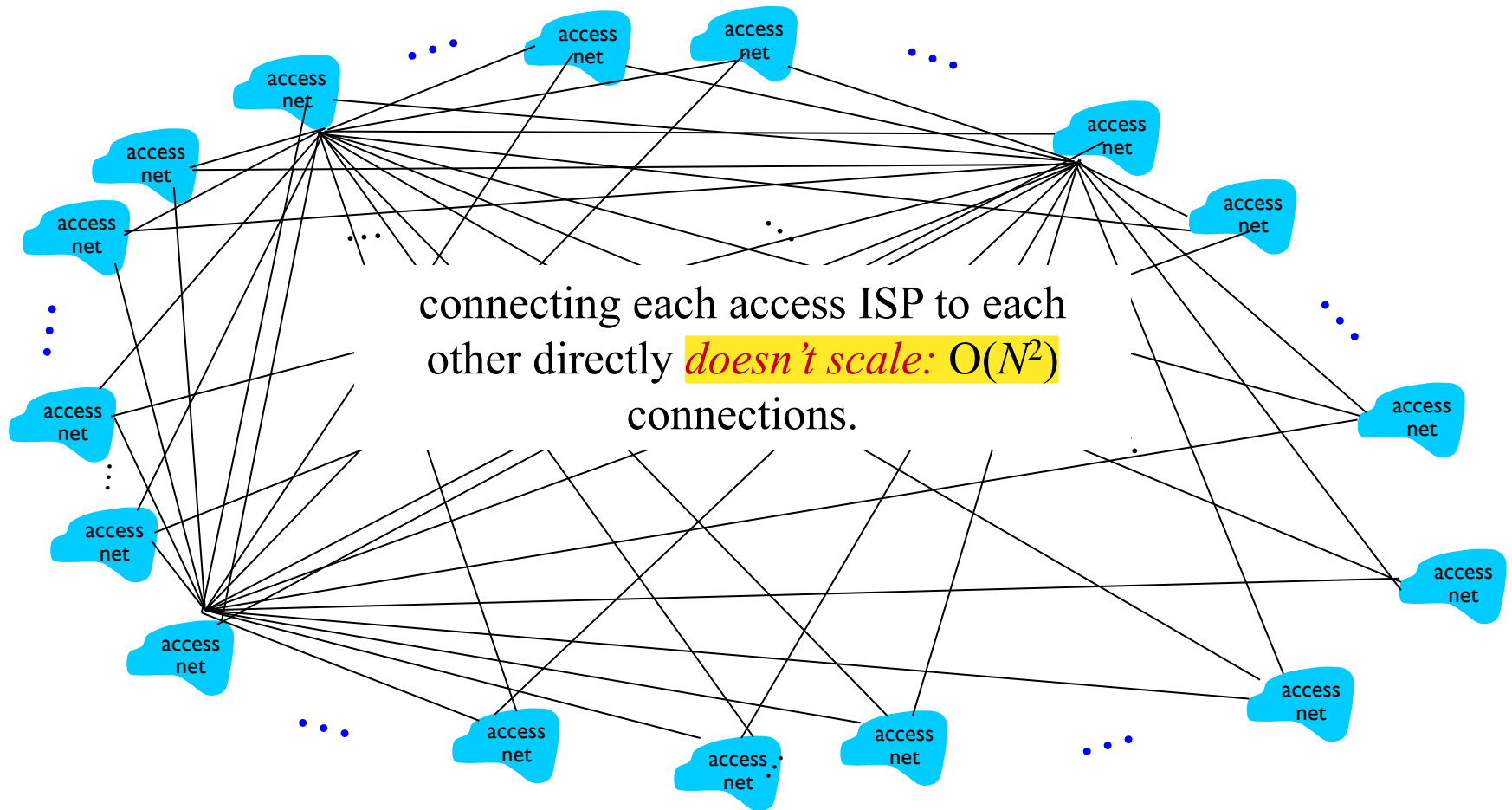
Internet structure: network of networks

Question: given *millions* of access ISPs, how to connect them together?



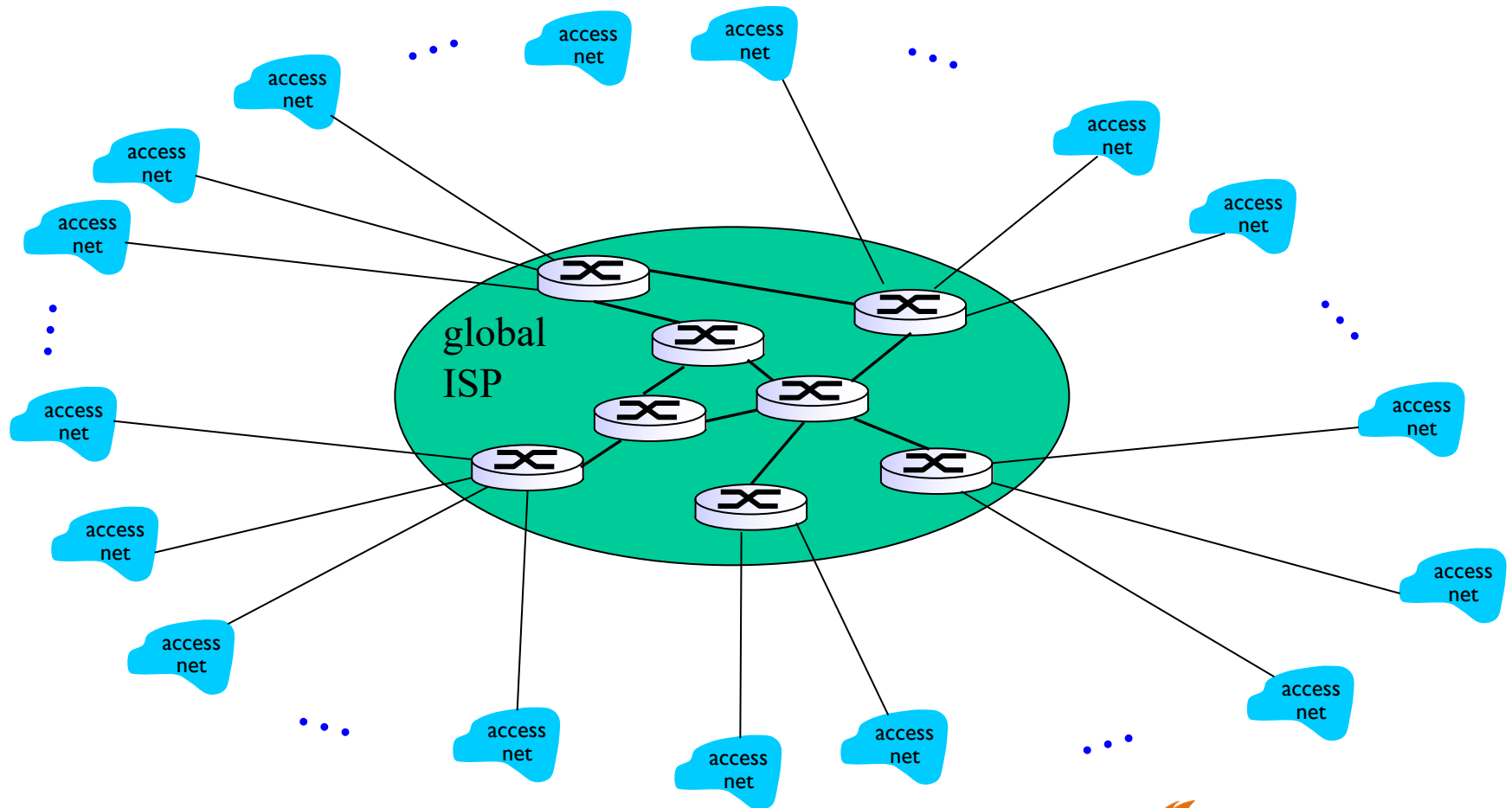
Internet structure: network of networks

Option 1: connect each access ISP to every other access ISP?



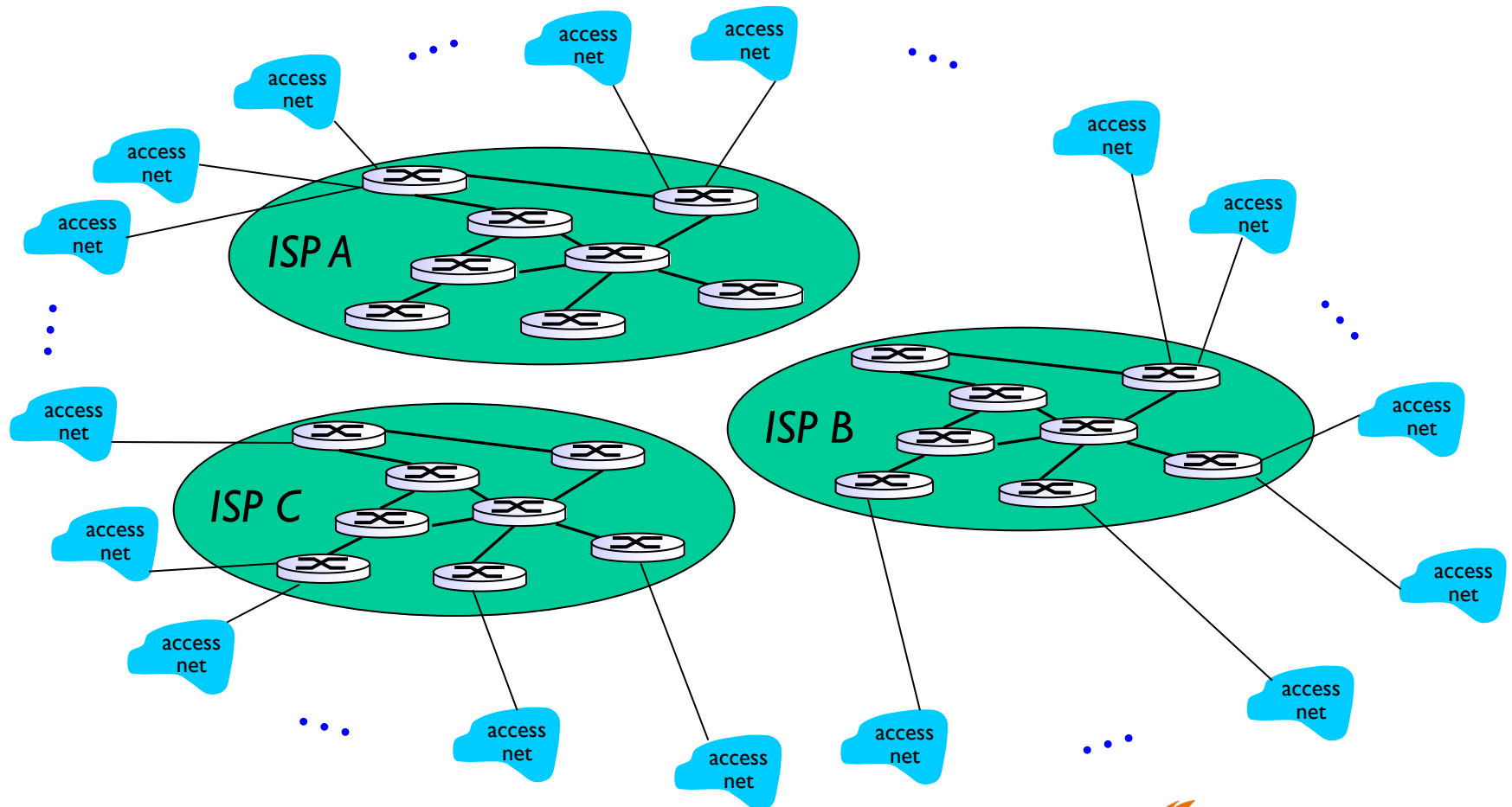
Internet structure: network of networks

Option 2: connect each access ISP to a global transit ISP? **Customer** and **provider** ISPs have economic agreement.



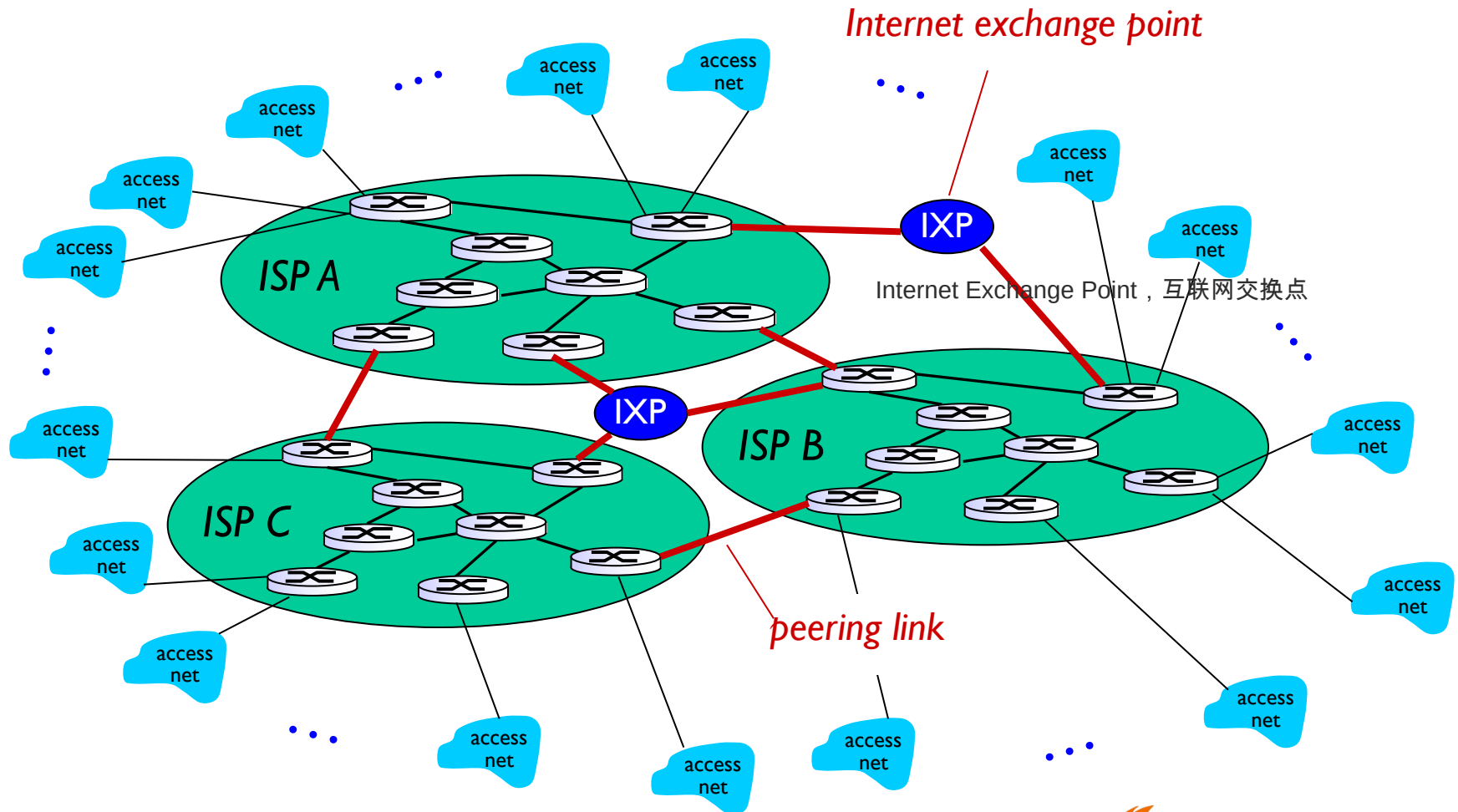
Internet structure: network of networks

But if one global ISP is viable business, there will be competitors



Internet structure: network of networks

But if one global ISP is viable business, there will be competitors which must be interconnected



Internet structure: network of networks

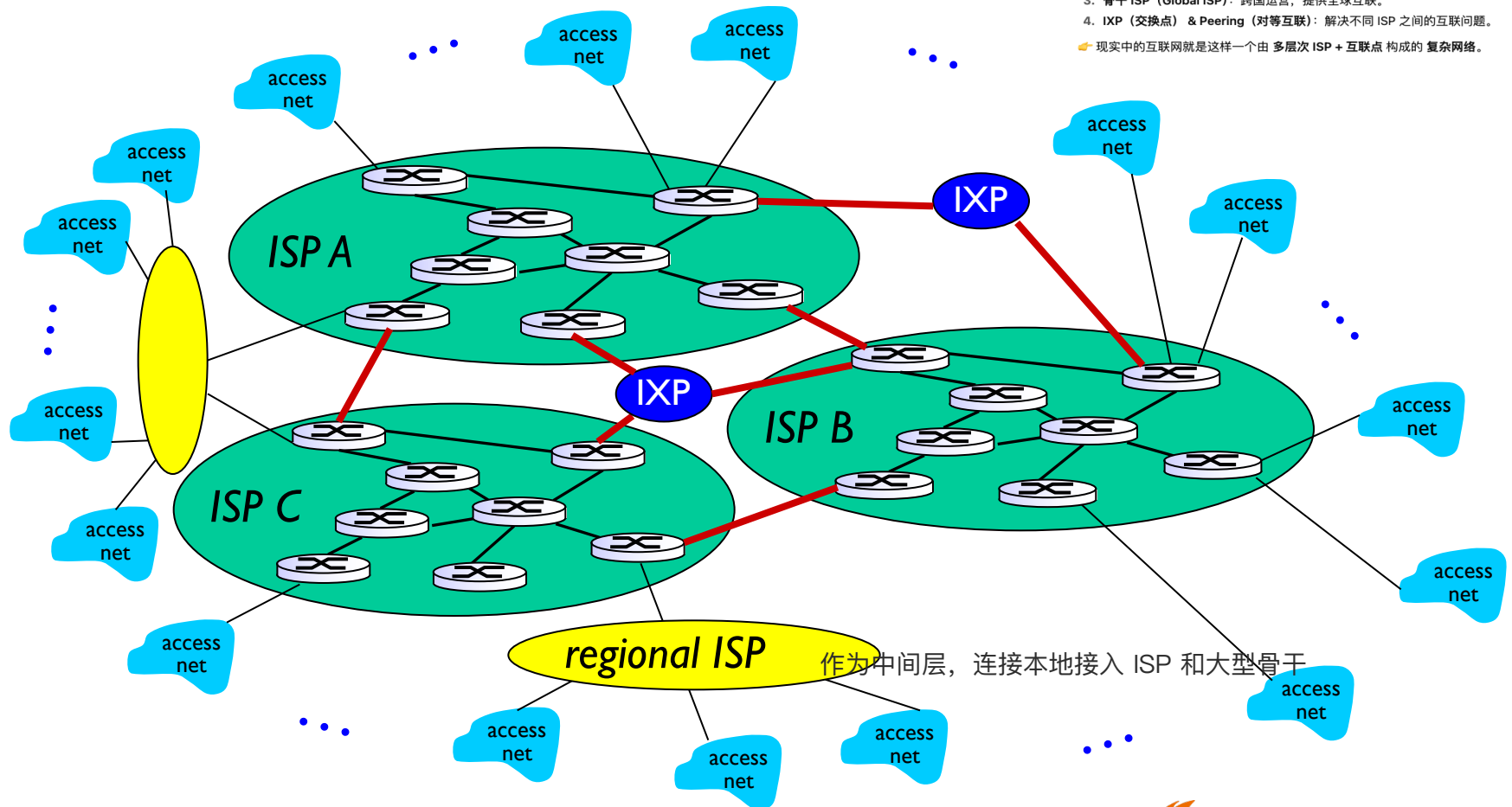
... and regional networks may arise to connect access nets to ISPs

总结

互联网的层次结构:

1. 接入 ISP (Access ISP): 家庭、学校、公司直接连的网络。
2. 区域 ISP (Regional ISP): 把多个接入 ISP 聚合起来, 接入更大网络。
3. 骨干 ISP (Global ISP): 跨国运营, 提供全球互联。
4. IXP (交换点) & Peering (对等互联): 解决不同 ISP 之间的互联问题。

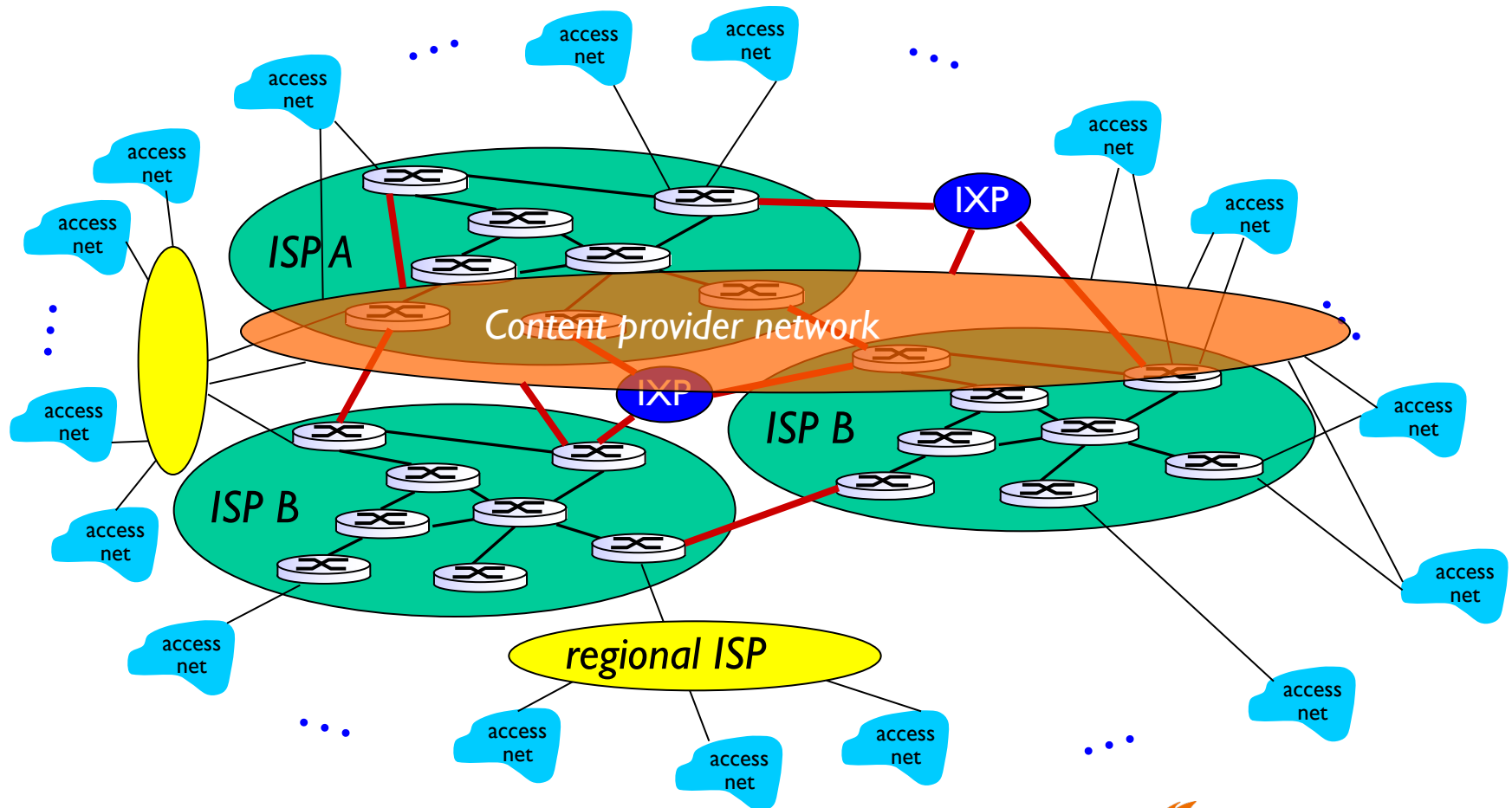
现实中的互联网就是这样一个由多层次 ISP + 互联点 构成的复杂网络。



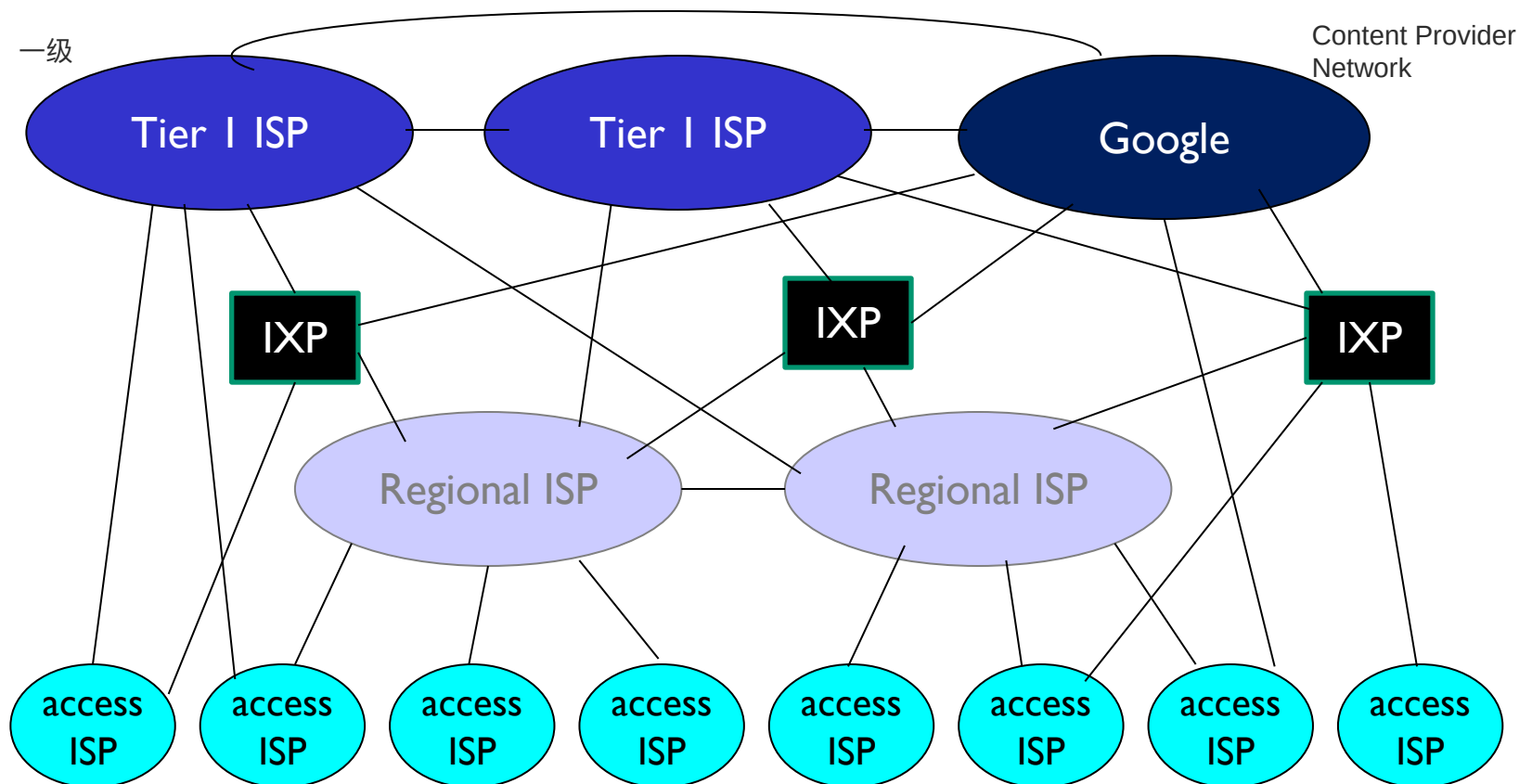
作为中间层, 连接本地接入 ISP 和大型骨干

Internet structure: network of networks

... and **content provider networks** (e.g., Google, Microsoft) may run their own network, to bring services, content close to end users



Internet structure: network of networks



- “Tier-1” commercial ISPs (e.g., Level 3, Sprint, AT&T, NTT), national & international coverage
- **Multi-homing**: to connect to two or more provider ISPs 一个 ISP 同时连到多个上级 ISP, 提高可靠性
- **Peer**: a pair of nearby ISPs **at the same level**
- **Content provider network** (e.g., Google): private network that connects its data centers to Internet, often bypassing tier-1, regional ISPs

Chapter 1: roadmap

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1.6 networks under attack: security

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Performance Metric (性能指标)

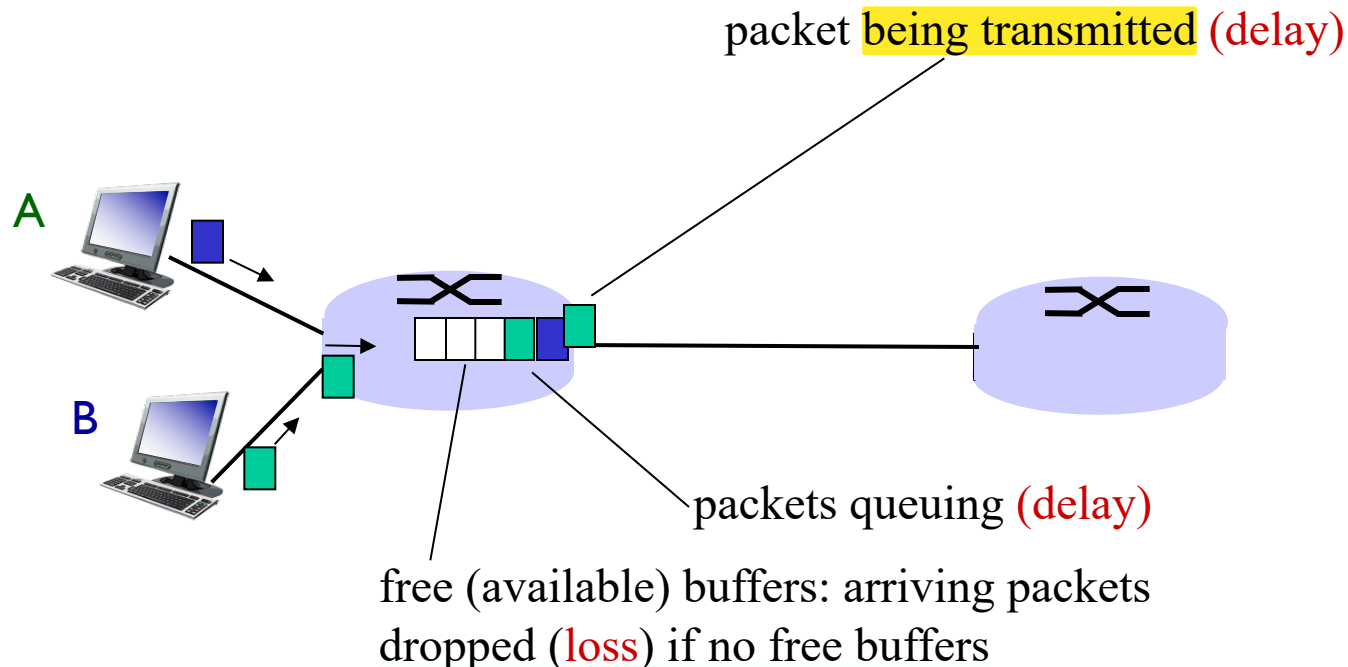
We would like Internet services to be able to **move as much data** as we want **between any two end systems**, instantaneously, **without any loss of data**.

- ❖ Delay (时延)
 - Nodal delay; end-to-end delay
- ❖ Packet loss (丢包)
- ❖ Throughput (吞吐量): the amount of data per second that can be transferred between end systems

How do loss and delay occur?

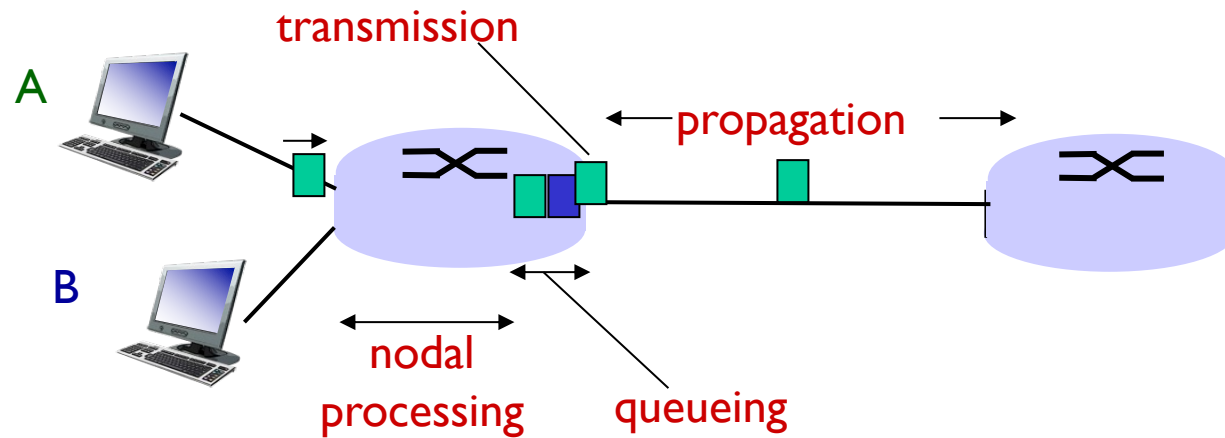
Packets **queue** in router buffers

- ❖ packet arrival rate to link (temporarily) exceeds output link capacity
- ❖ packets queue, wait for turn

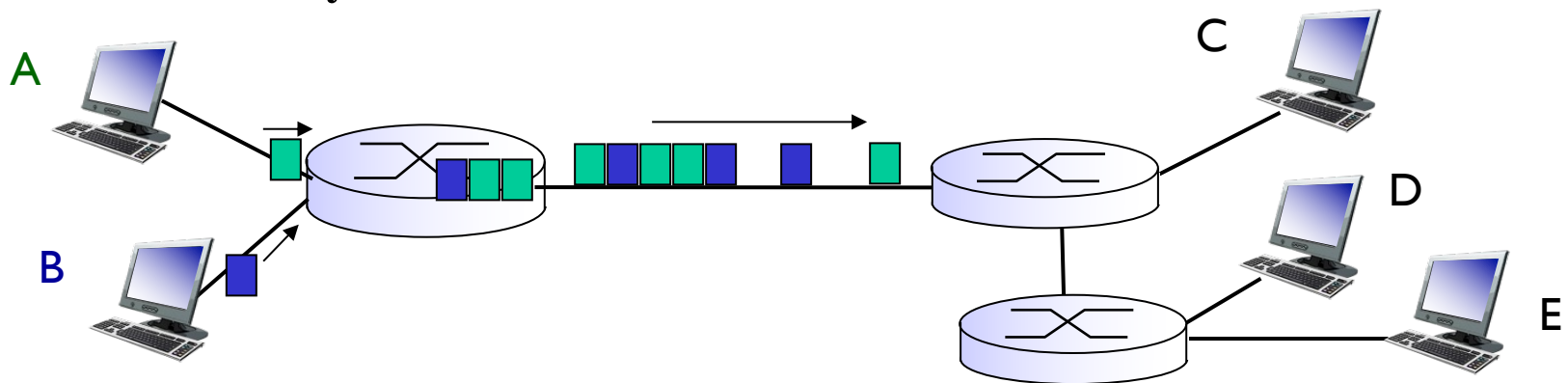


Nodal vs End-to-End Delay

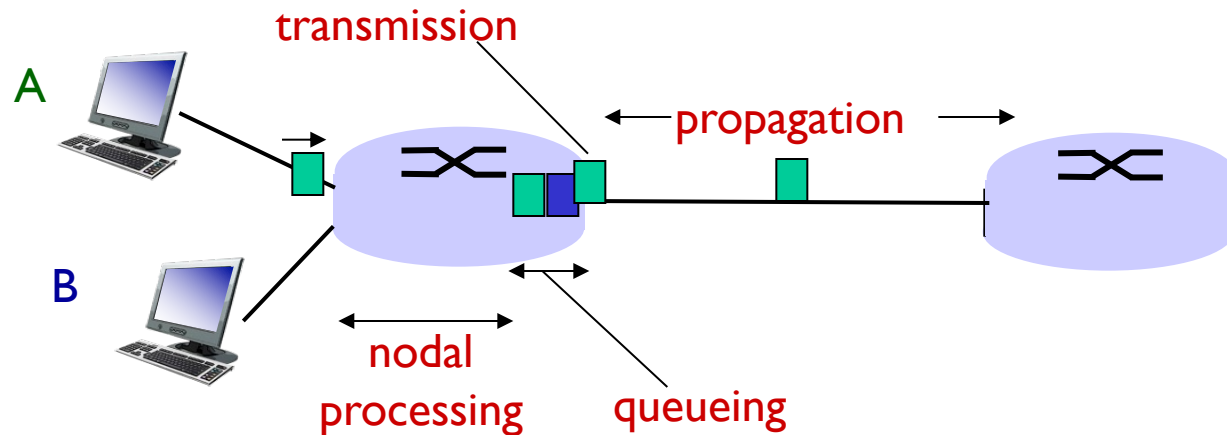
Nodal (节点) delay



End-to-end delay



Four sources of nodal delay



Nodal (节点) delay

$$d_{\text{nodal}} = d_{\text{proc}} + d_{\text{queue}} + d_{\text{trans}} + d_{\text{prop}}$$

d_{proc} : nodal processing

- check bit errors
- determine output link
- typically < msec

路由器检查分组的 首部 (header):

检查比特错误 (check bit errors)

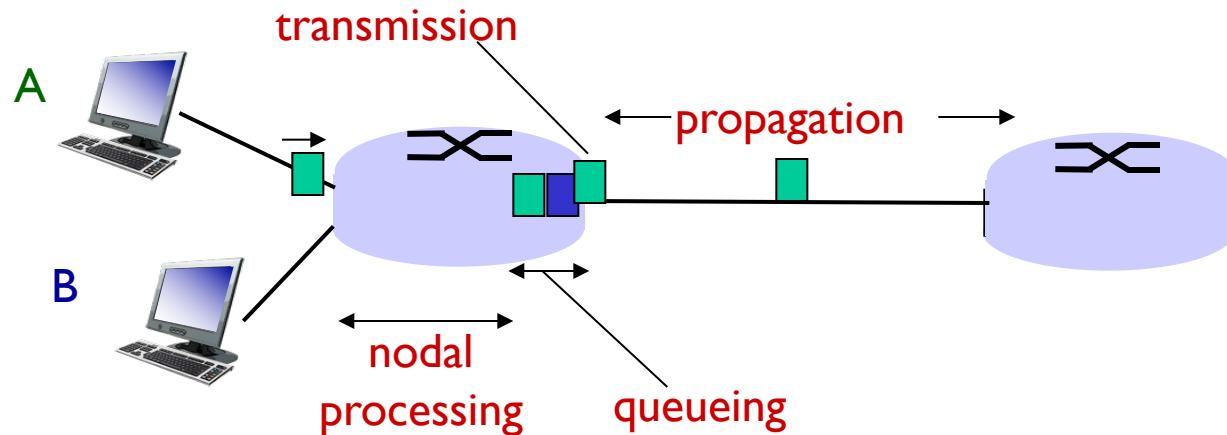
确定下一跳的输出链路 (determine output link)

通常非常快 (小于 1 毫秒) 。

d_{queue} : queueing delay

- time waiting at output link for transmission
- depends on congestion level of router

Four sources of nodal delay



Nodal (节点) delay

$$d_{\text{nodal}} = d_{\text{proc}} + d_{\text{queue}} + d_{\text{trans}} + d_{\text{prop}}$$

把一个分组 推出链路 所需的时间

d_{trans} : transmission delay (传输时延):

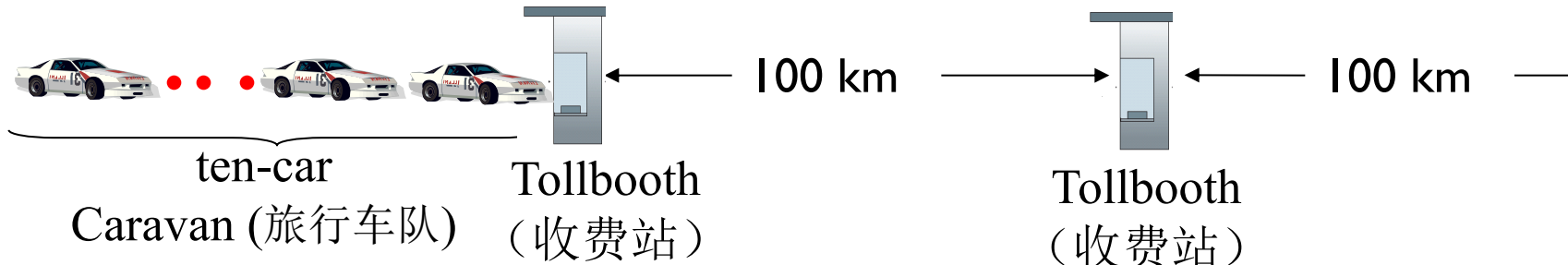
- L : packet length (bits)
- R : transmission rate (bps)
- $d_{\text{trans}} = L/R$

d_{prop} : propagation delay (传播时延):

- d : length of physical link
- s : propagation speed in medium ($\sim 2 \times 10^8$ m/sec)
- $d_{\text{prop}} = d/s$

d_{trans} and d_{prop}
very different

Caravan analogy

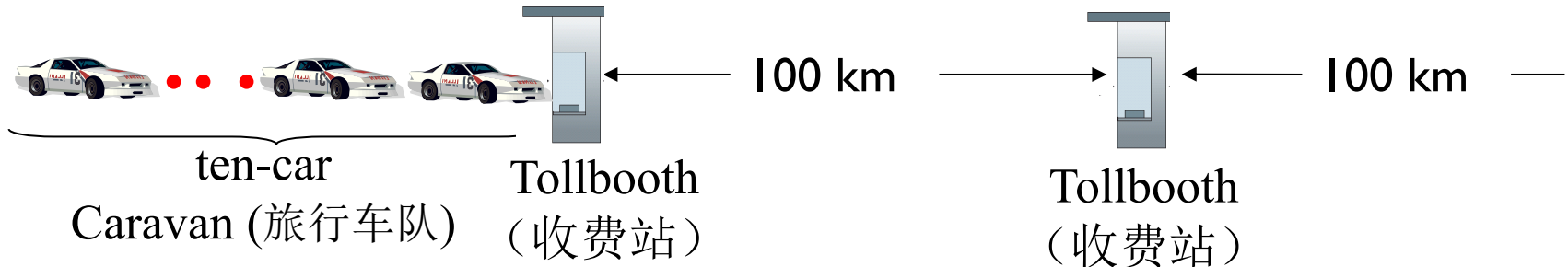


- ❖ Car $\sim$ bit; caravan $\sim$ packet
- ❖ Tollbooth takes 12 sec to service one car $\sim$ transmission time
- ❖ Cars “propagate” at 100 km/hr $\sim$ propagation speed

Q: How long does it take the last car to arrive at the 2nd tollbooth?

- **Transmission delay**: time to “push” entire caravan through tollbooth onto highway = $12 \times 10 = 120$ sec
- **Propagation delay**: time for last car to propagate from 1st to 2nd toll both: $100\text{km}/(100\text{km/hr}) = 1$ hr
- *A:* 62 minutes

Caravan analogy (more)



- ❖ Suppose cars now “propagate” at 1000 km/hr
- ❖ and suppose tollbooth now takes one min to service a car
- ❖ **Q: How long does it take the last car to arrive at the 2nd tollbooth?**
 - Transmission delay: $1 \text{ min} * 10 = 10 \text{ min}$
 - Propagation delay: $100 \text{ km} / 1000 \text{ km/hr} = 6 \text{ min}$

After 7 min, 1st car arrives at second booth; three cars still at 1st booth.

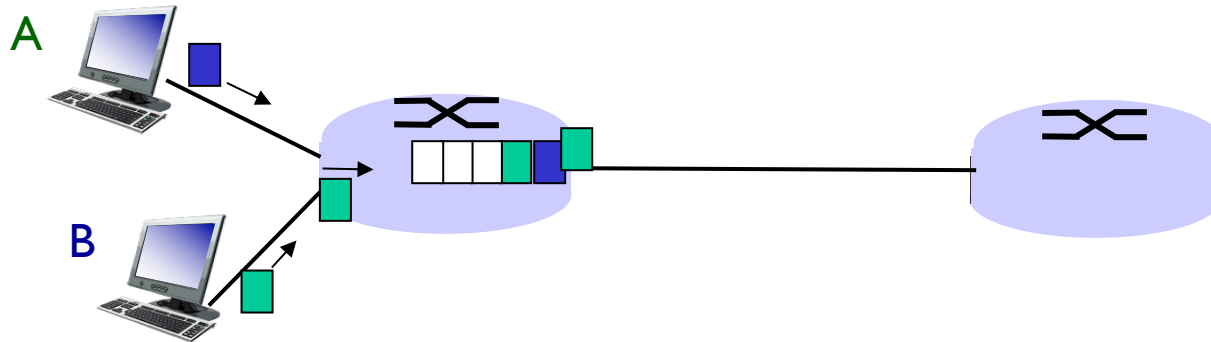
[Transmission versus Propagation Delay Interactive Animation \(unicam.it\)](http://unicam.it)

Queueing delay

Nodal (节点) delay

$$d_{\text{nodal}} = d_{\text{proc}} + d_{\text{queue}} + d_{\text{trans}} + d_{\text{prop}}$$

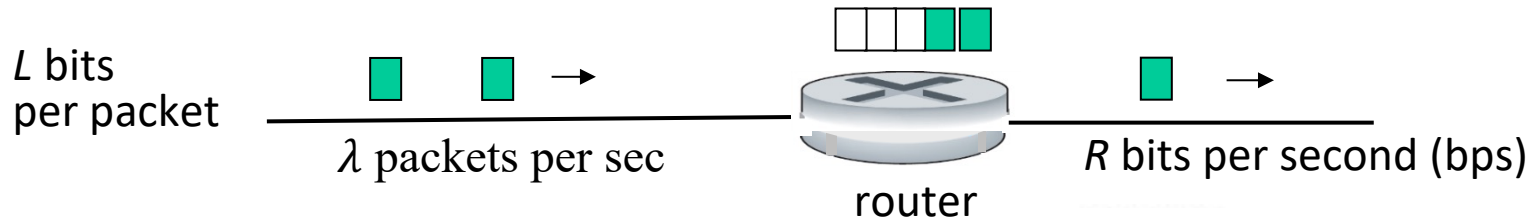
Unlike the other three delays (namely, d_{proc} , d_{trans} , d_{prop}), the queuing delay can vary from packet to packet.



When characterizing queuing delay, statistical measures:

- average queuing delay
- variance of queuing delay
- the probability that the queuing delay exceeds some specified value

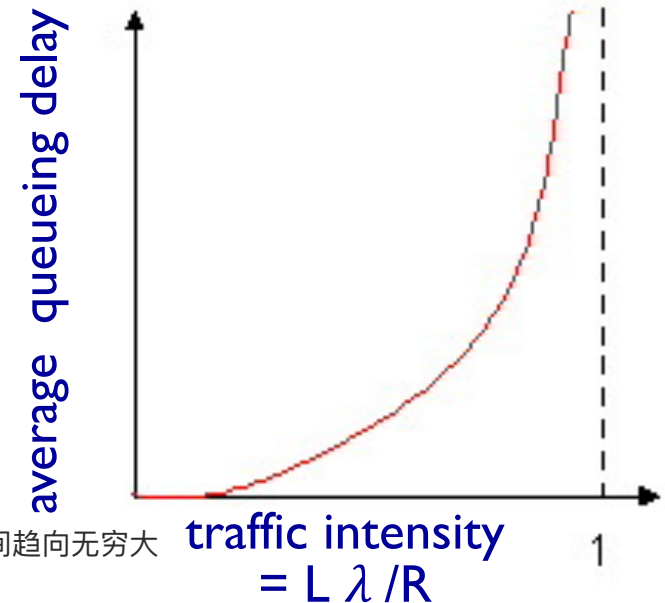
Queueing delay at a router



- ❖ L : packet length (bits)
- ❖ λ : average packet arrival rate (packets per sec)
- ❖ R : transmission rate (bps)

Traffic intensity = $L \lambda / R$

- ❖ $L \lambda / R \rightarrow 0$: avg. queueing delay small
- ❖ $L \lambda / R \rightarrow 1$: avg. queueing delay $\rightarrow \infty$ (queueing theory) 随机性叠加后, 队伍会越积越长, 平均等待时间趋向无穷大
- ❖ $L \lambda / R > 1$: more “work” arriving than can be serviced, average delay infinite!



这就是概率论里的结果:

- 在 M/M/1 排队模型中 (单服务器, 泊松到达, 指数分布服务时间):

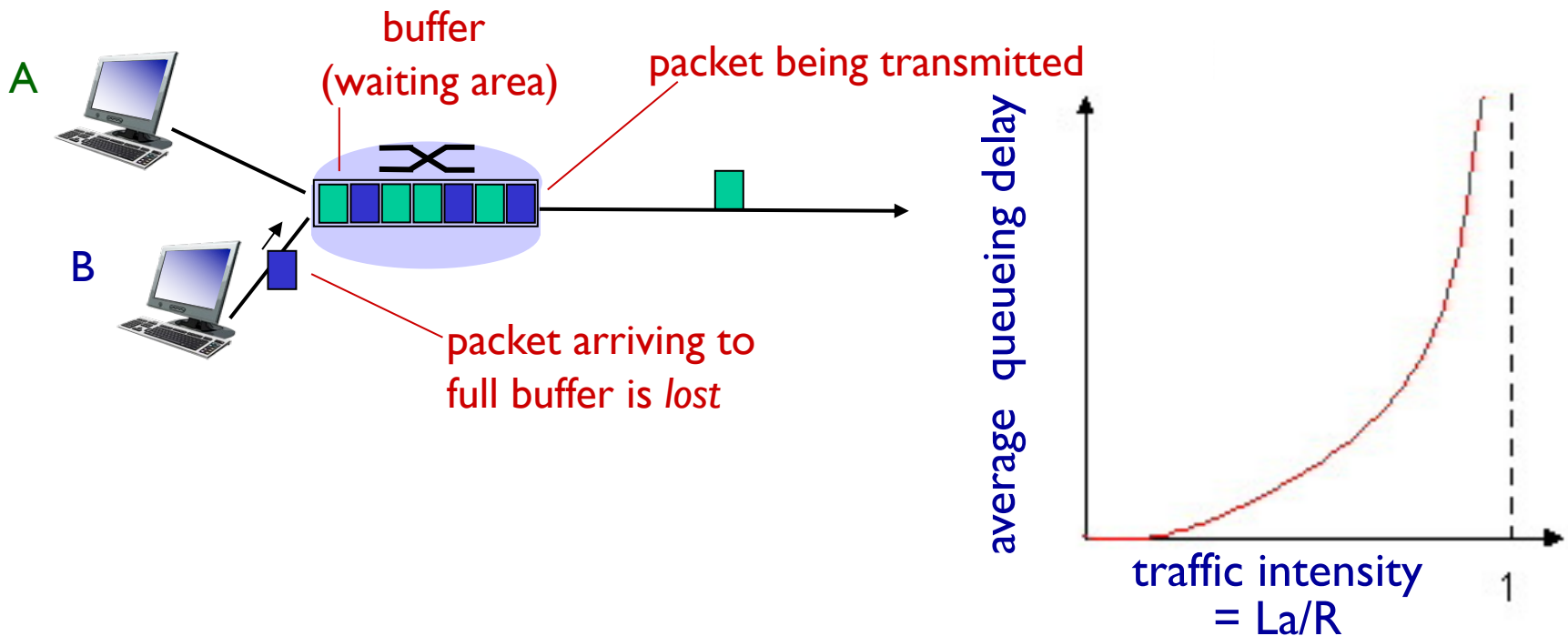
$$\text{平均等待时间} = \frac{\rho}{1 - \rho} \cdot \frac{1}{\mu}$$

当 $\rho \rightarrow 1$ 时, 分母 $(1 - \rho) \rightarrow 0$, 等待时间 $\rightarrow \infty$ 。

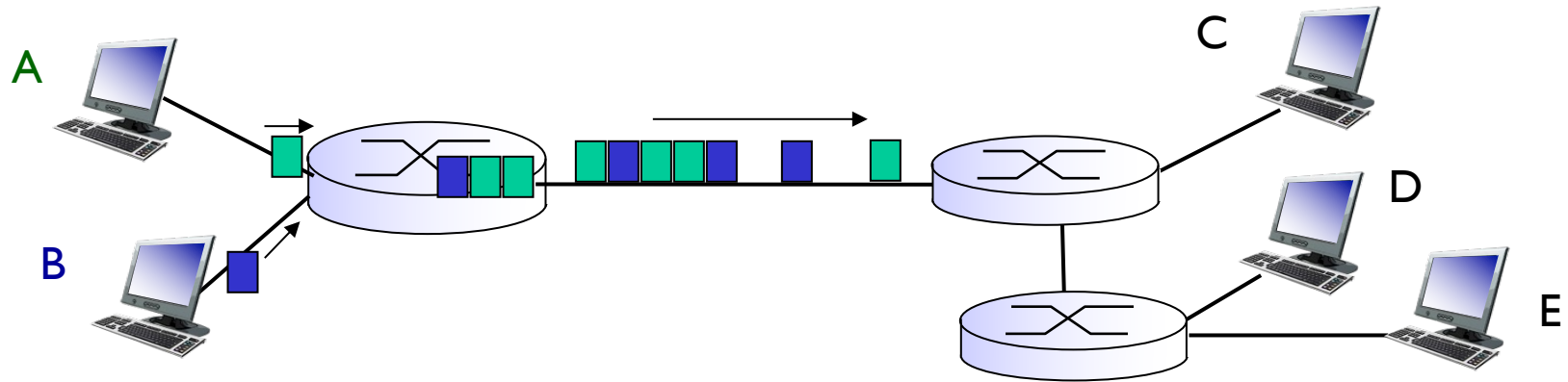
[Queuing and Loss Interactive Animation \(unicam.it\)](http://unicam.it)

Packet loss

- ❖ Queue (aka buffer) preceding link in buffer has finite capacity
- ❖ Packet arriving to full queue will be **dropped** (aka lost)
- ❖ Lost packet may be **retransmitted** by previous node, by source end system, or not at all



End-to-End Delay



Suppose d_{queue} is negligible; $N-1$ routers between two hosts

End-to-End delay

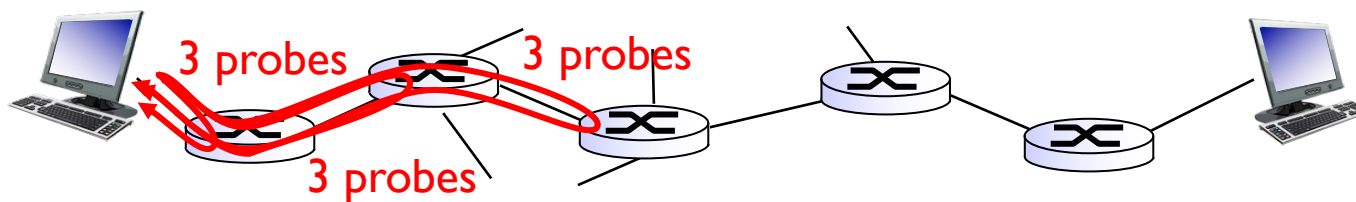
$$d_{\text{end-end}} = N (d_{\text{proc}} + d_{\text{trans}} + d_{\text{prop}})$$

However, queuing delay can be significant and time-varying

“Real” Internet delays and routes

- ❖ What do “real” Internet delay & loss look like?
- ❖ **Traceroute** program: provides delay measurement from source to router along end-end Internet path towards destination. For all i :
 - sends three packets with a **time-to-live (TTL)** of i ; will reach router i on path towards destination
 - router i will return packets to sender
 - sender times interval between transmission and reply.

每经过一个路由器, TTL 会减 1。



如果 TTL 变成 0, 该路由器会丢弃分组并返回一个 ICMP 报文 ("Time Exceeded")

$N-1$ routers, send $3N$ packets

“Real” Internet delays, routes

traceroute: gaia.cs.umass.edu to www.eurecom.fr

3 round-trip delay measurements from
gaia.cs.umass.edu to cs-gw.cs.umass.edu

左边是跳数

1	cs-gw (128.119.240.254)	1 ms	1 ms	2 ms
2	border1-rt-fa5-1-0.gw.umass.edu (128.119.3.145)	1 ms	1 ms	2 ms
3	cht-vbns.gw.umass.edu (128.119.3.130)	6 ms	5 ms	5 ms
4	jn1-at1-0-0-19.wor.vbns.net (204.147.132.129)	16 ms	11 ms	13 ms
5	jn1-so7-0-0-0.wae.vbns.net (204.147.136.136)	21 ms	18 ms	18 ms
6	abilene-vbns.abilene.ucaid.edu (198.32.11.9)	22 ms	18 ms	22 ms
7	nycm-wash.abilene.ucaid.edu (198.32.8.46)	22 ms	22 ms	22 ms
8	62.40.103.253 (62.40.103.253)	104 ms	109 ms	106 ms
9	de2-1.de1.de.geant.net (62.40.96.129)	109 ms	102 ms	104 ms
10	de.fr1.fr.geant.net (62.40.96.50)	113 ms	121 ms	114 ms
11	renater-gw.fr1.fr.geant.net (62.40.103.54)	112 ms	114 ms	112 ms
12	nio-n2.cssi.renater.fr (193.51.206.13)	111 ms	114 ms	116 ms
13	nice.cssi.renater.fr (195.220.98.102)	123 ms	125 ms	124 ms
14	r3t2-nice.cssi.renater.fr (195.220.98.110)	126 ms	126 ms	124 ms
15	eurecom-valbonne.r3t2.ft.net (193.48.50.54)	135 ms	128 ms	133 ms
16	194.214.211.25 (194.214.211.25)	126 ms	128 ms	126 ms
17	***			
18	***			
19	fantasia.eurecom.fr (193.55.113.142)	132 ms	128 ms	136 ms

trans-oceanic link

不同探测包走的路径不一样 或者
网络抖动

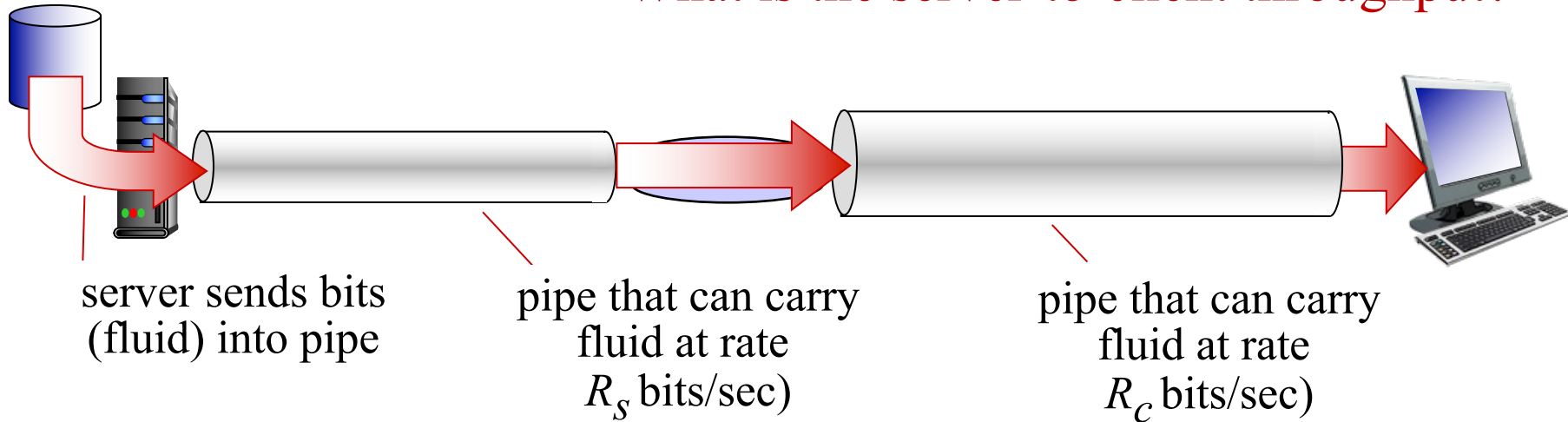
looks like delays
decrease! Why?

* means no response (probe lost, router not replying)

* Do some traceroutes from exotic countries at www.traceroute.org

Throughput

What is the server-to-client throughput?

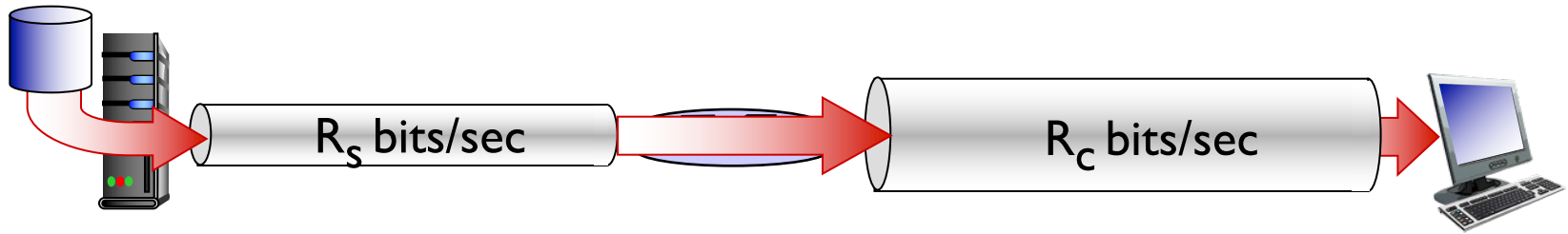


Throughput: rate (bits/time unit) at which bits transferred between sender/receiver

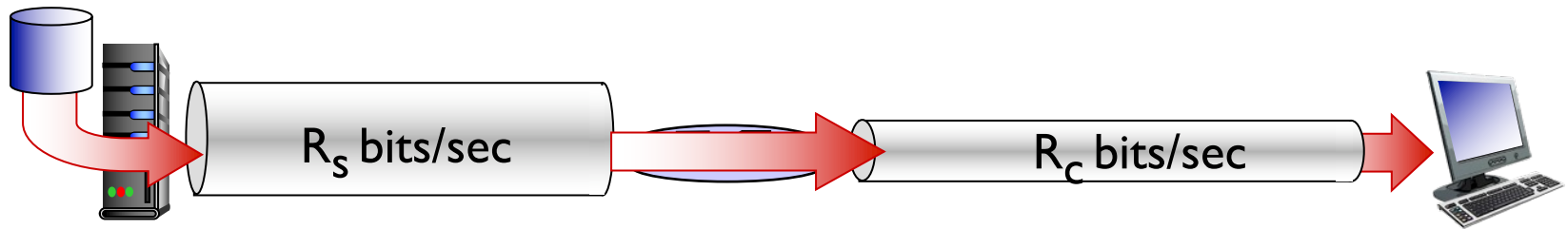
- ❖ *instantaneous*: rate at given point in time
- ❖ *average*: rate over longer period of time
 - A file of F bits; it takes T seconds for the receiver to receive the file → Average throughput of F/T bits/sec.

Throughput (more)

❖ $R_s < R_c$ What is average end-end throughput?



❖ $R_s > R_c$ What is average end-end throughput?

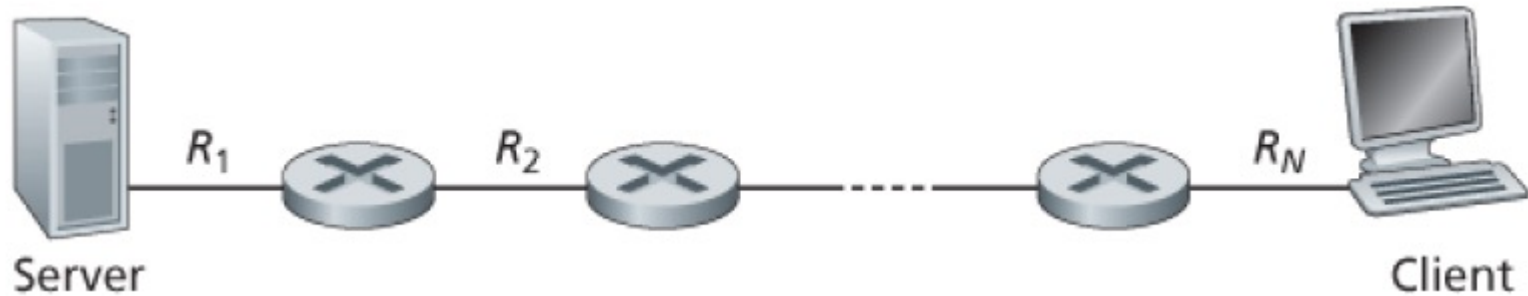


$$\min\{R_s, R_c\}$$

bottleneck (瓶颈) link

link on end-end path that constrains end-end throughput

Throughput (more)



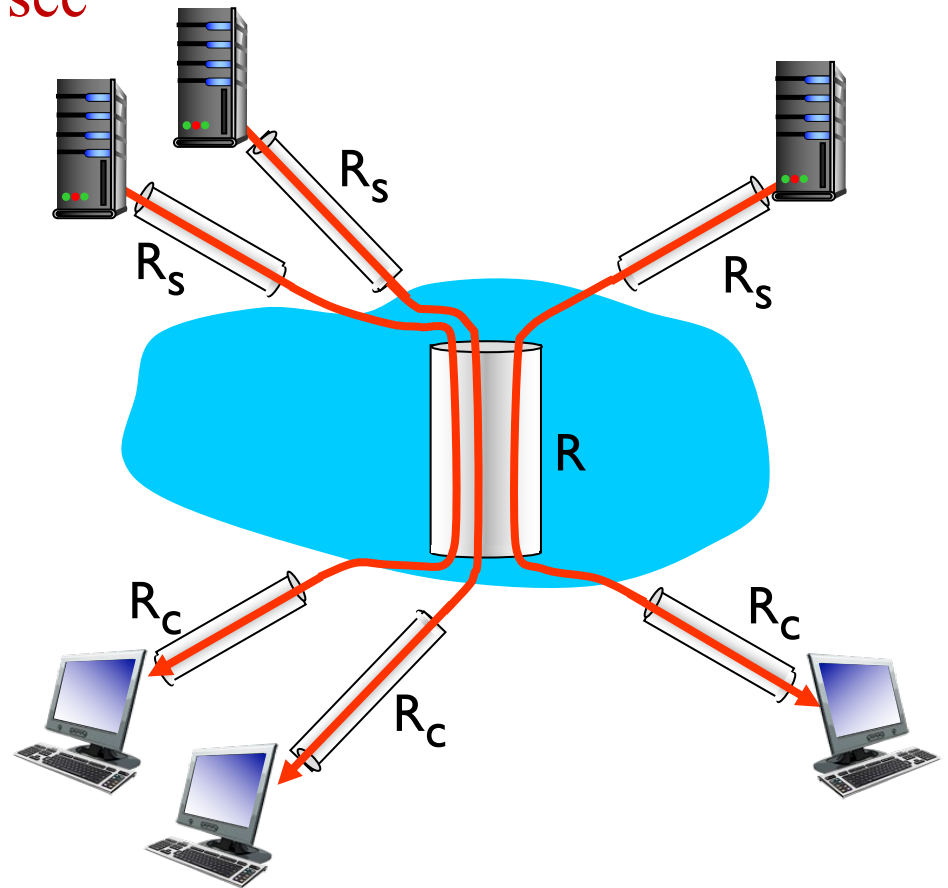
What is average end-end throughput?

$$\min\{R_1, R_2, \dots, R_N\}$$

Throughput: Internet scenario

10 connections (fairly) share
backbone bottleneck link R bits/sec

- ❖ Per-connection end-end throughput:
 $\min\{R_c, R_s, R/10\}$
- ❖ In practice: R_c or R_s is often bottleneck



Chapter 1: roadmap

1.1 what is the Internet?

1.2 network edge

- end systems, access networks, links

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- packet switching, circuit switching, network structure

1.4 delay, loss, throughput in networks

1.5 protocol layers, service models

1.6 networks under attack: security

1.7 history

Protocol “layers”

Networks are complex,
with many “pieces”:

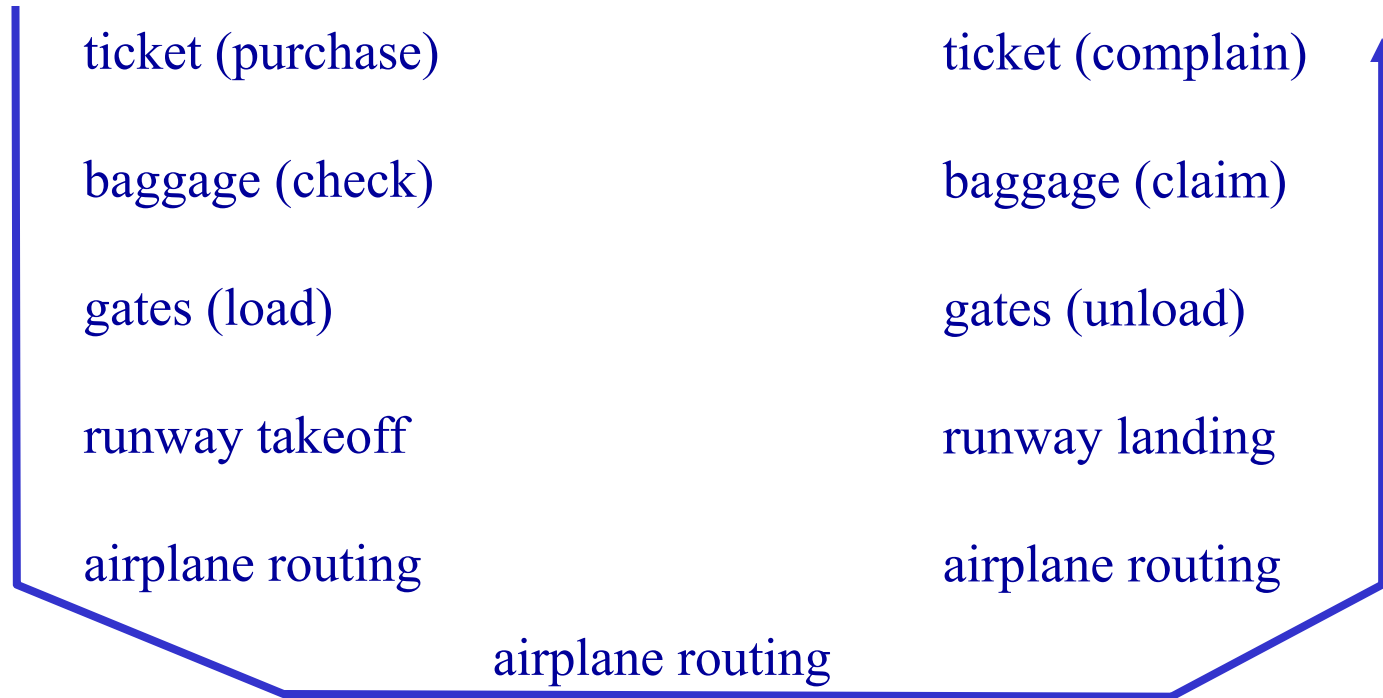
- hosts
- routers
- links of various media
- applications
- protocols
- hardware, software

Question:

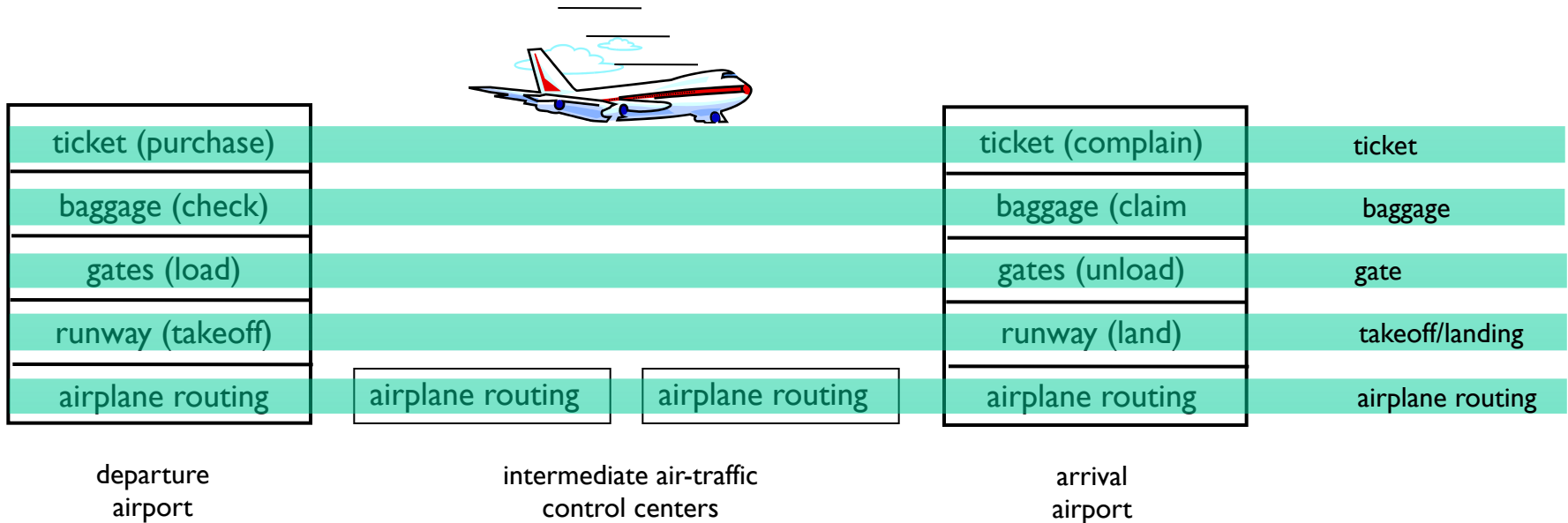
is there any hope of
organizing structure of
network?

.... or at least our
discussion of networks?

Organization of air travel



Layering of airline functionality



Layers: Each layer **provide** services **to the layer above**

- **via** its own internal-layer actions
- **relying on** services provided by layer below

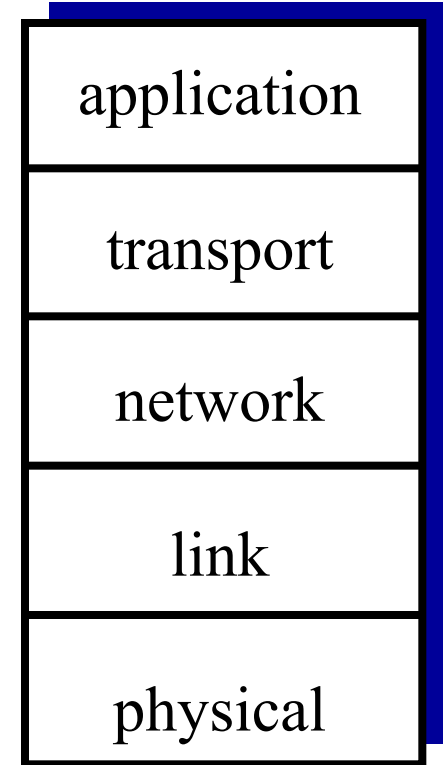
Why layering?

Dealing with complex systems:

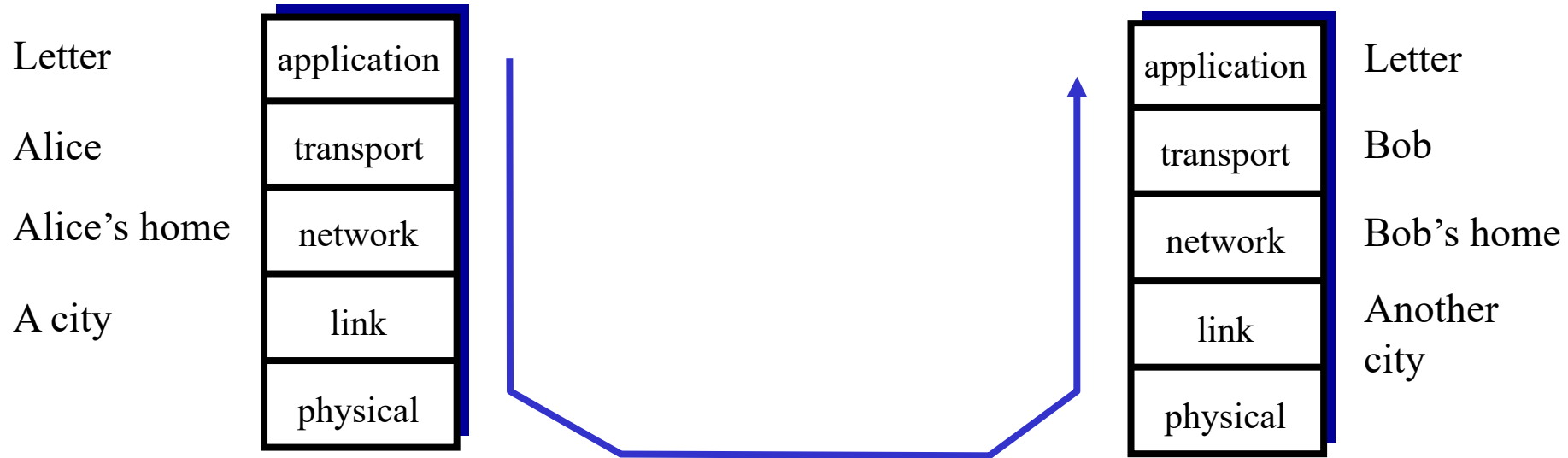
- ❖ Explicit structure allows identification, relationship of complex system's pieces
 - layered reference model for discussion
- ❖ modularization eases maintenance, updating of system
 - change of implementation of layer's service **transparent** to rest of system
 - e.g., change in gate procedure doesn't affect rest of system
- ❖ drawback?
 - One layer may duplicate lower layer functionality
 - Functionality at one layer may need information that is present only in another layer

Internet protocol stack

- ❖ **Application:** supporting network applications
 - FTP, SMTP, HTTP
- ❖ **Transport:** process (进程)-process data transfer
 - TCP, UDP
- ❖ **Network:** routing of datagrams from source host to destination host
 - IP, routing protocols
- ❖ **Link:** data transfer between neighboring network elements
 - Ethernet, 802.111 (WiFi), PPP
- ❖ **Physical:** bits “on the wire”



Layering of functionality



How the mail is delivered (e.g., through highway or rail) between cities

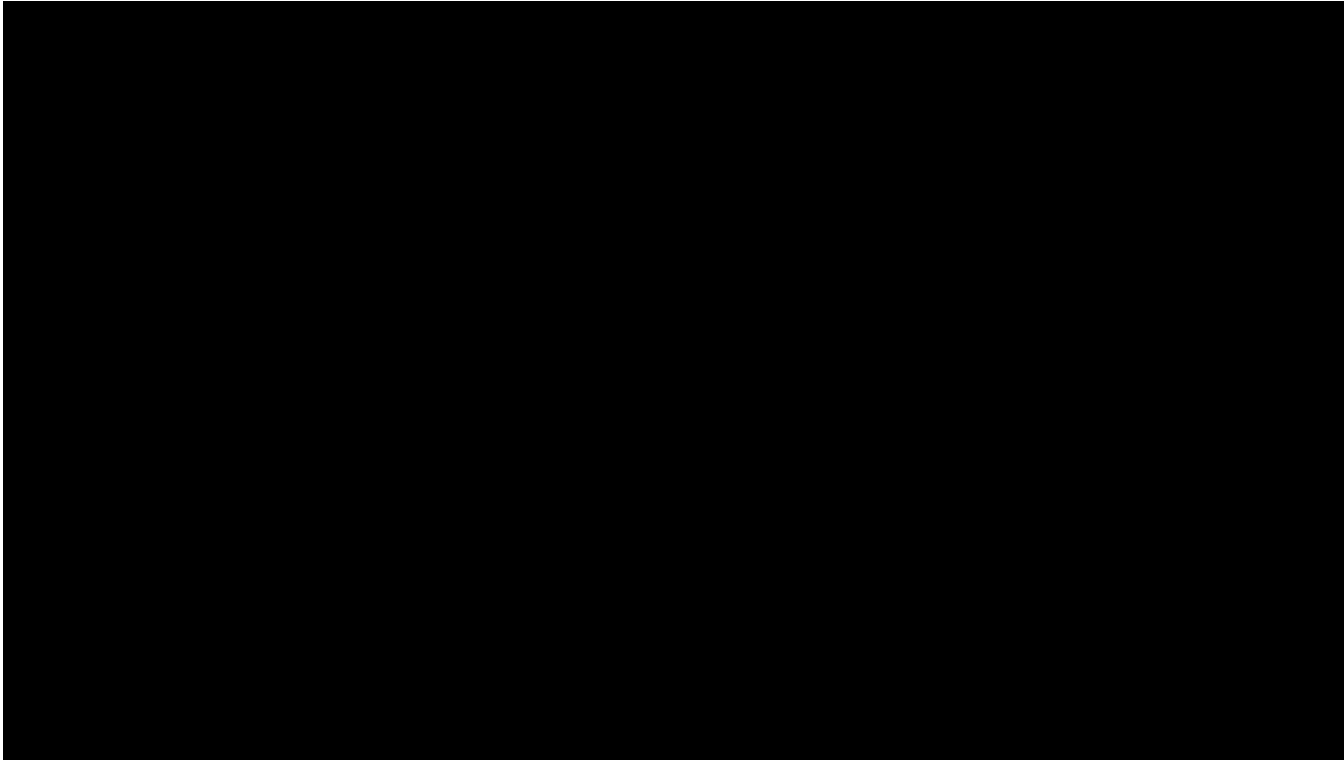
Layers: Each layer **provide** services **to the layer above**

- **via** its own internal-layer actions
- **relying on** services provided by layer below

For example, **reliable delivery** of messages at layer n :

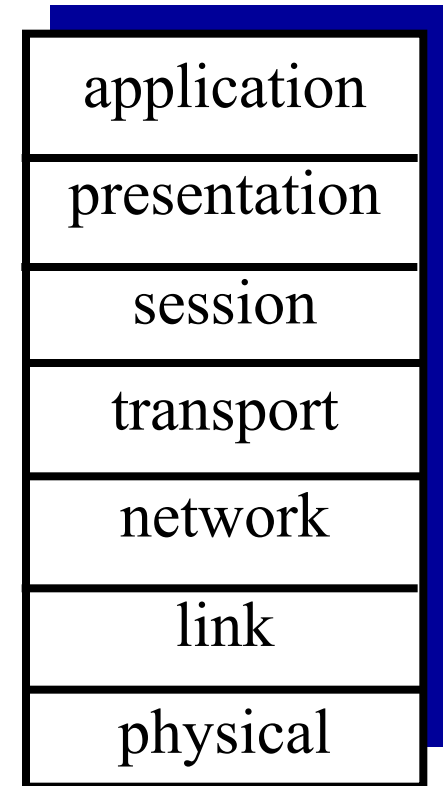
- Adding layer n functionality to **detect and retransmit lost messages**
- Using an **unreliable message delivery** service at layer $n-1$

Internet protocol stack

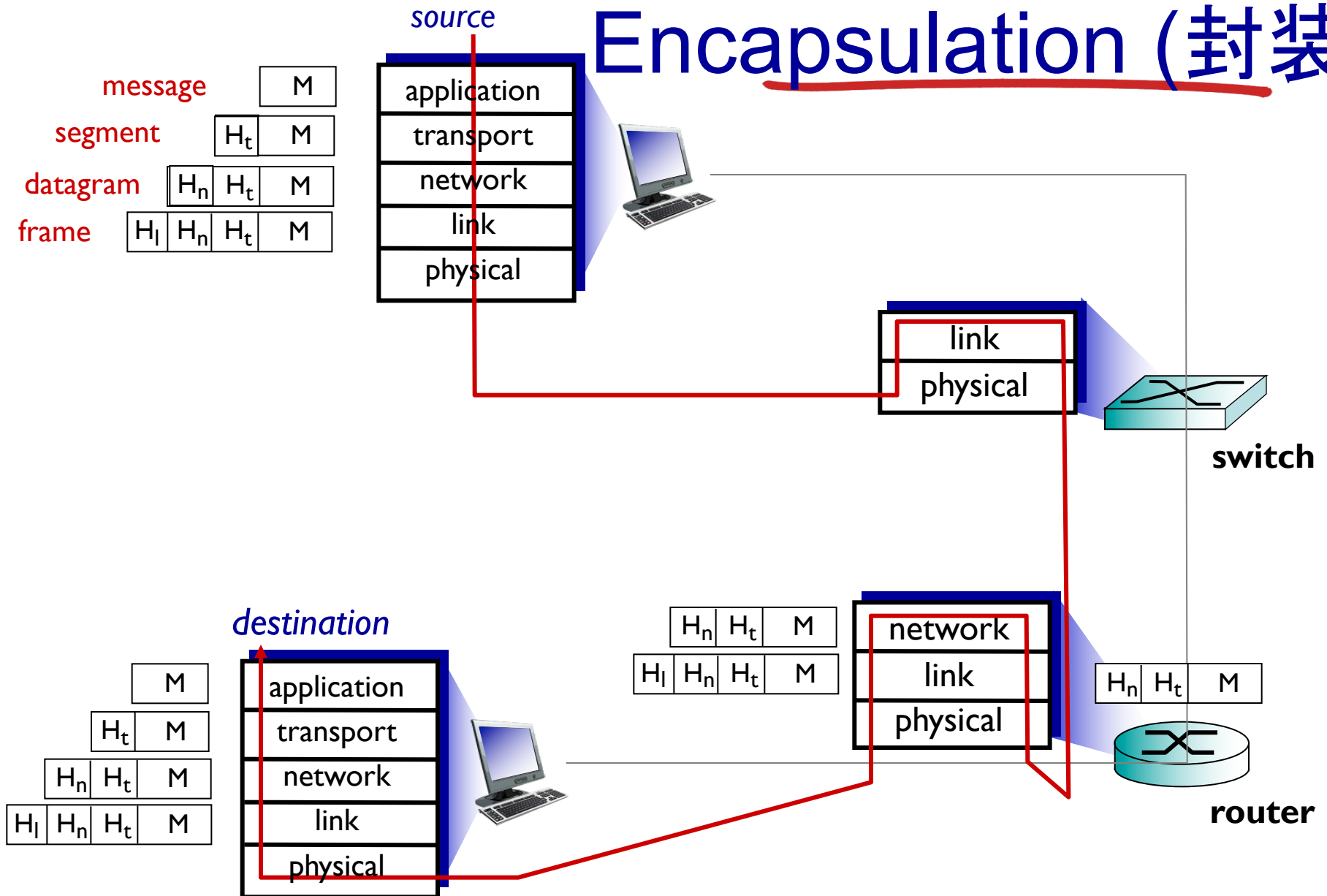


ISO/OSI reference model

- ❖ **Presentation:** allow applications to interpret meaning of data, e.g., encryption, compression, machine-specific conventions
- ❖ **Session:** synchronization, checkpointing, recovery of data exchange
- ❖ Internet stack “missing” these layers!
 - these services, *if needed*, must be implemented in **application**
 - needed?



Encapsulation (封装)



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Network security

We attach devices to the Internet because we want to **receive/send data** from/to the Internet

- ❖ **Field of network security:**
 - how bad guys can attack computer networks
 - how we can defend networks against attacks
 - how to design architectures that are immune to attacks
- ❖ **Internet not originally designed with (much) security in mind**
 - *original vision*: “a group of mutually trusting users attached to a transparent network” 😊
 - Internet protocol designers playing “catch-up”
 - security considerations in all layers!

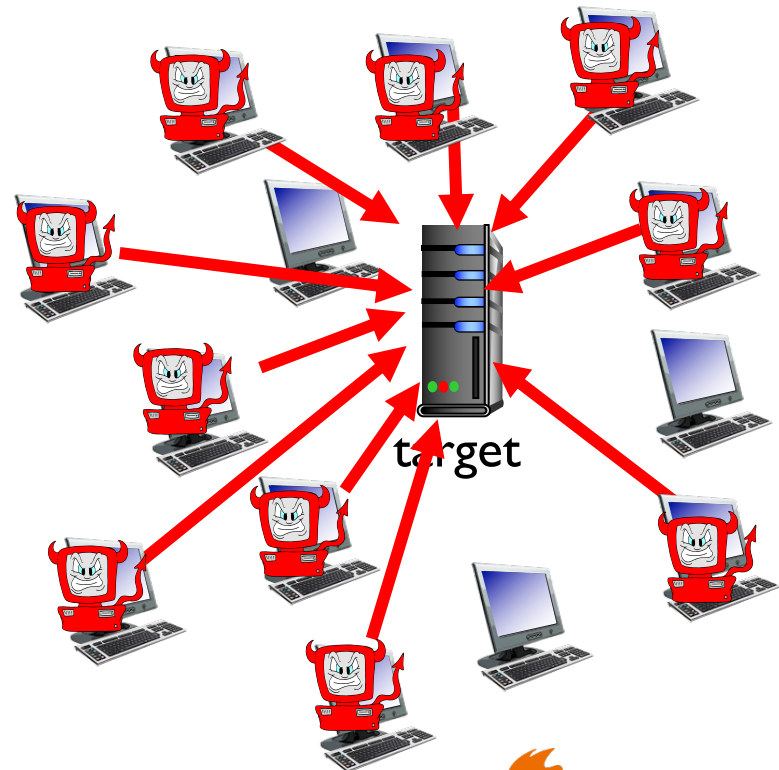
Bad guys: put malware (恶意软件) into hosts via Internet

- ❖ malware can get in host from:
 - *virus*: self-replicating infection by receiving/executing object (e.g., e-mail attachment) – **with** user interaction
 - *worm*: self-replicating infection by passively receiving object that gets itself executed – **without** user interaction
- ❖ **spyware malware** can record keystrokes, web sites visited, upload info to collection site
- ❖ infected host can be enrolled in **botnet** (僵尸网络), used for spam. DDoS attacks

Bad guys: attack server, network infrastructure

Denial of Service (DoS): attackers make resources (server, bandwidth) unavailable to legitimate (合法的) traffic by **overwhelming resource** with bogus (伪造的) traffic

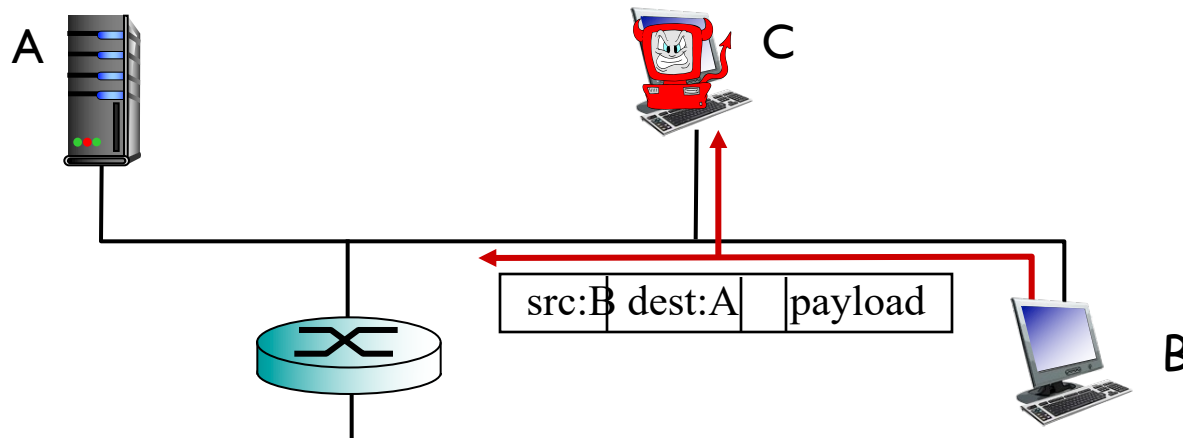
1. select target
2. break into hosts around the network (see botnet)
3. send packets to target from compromised hosts



Bad guys can sniff packets

Packet “sniffing”:

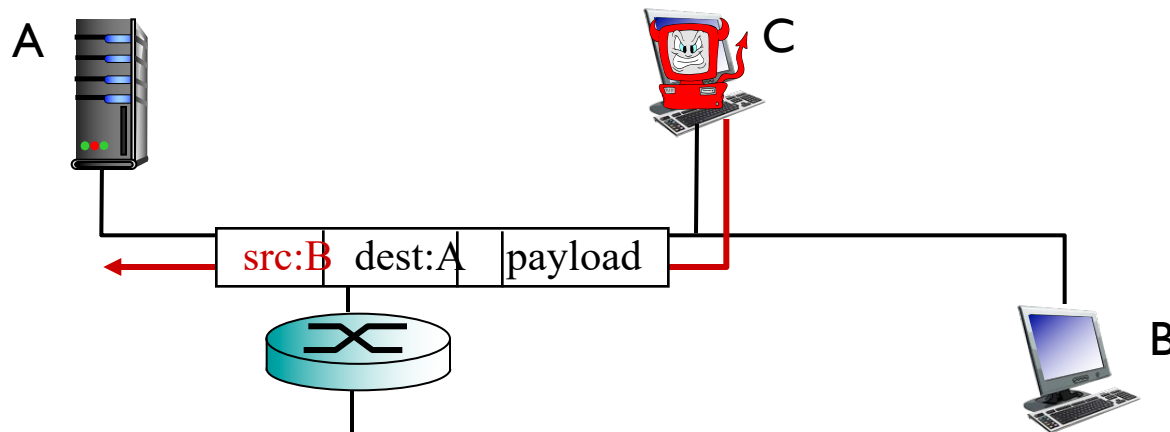
- Broadcast media (shared ethernet, wireless)
- Reads/records all packets (e.g., including passwords!) passing by
- They are difficult to detect



- ❖ wireshark software used for end-of-chapter labs is a (free) packet-sniffer

Bad guys can use fake addresses

IP spoofing: send packet with false source address



Lines of defense:

- **authentication:** proving you are who you say you are
 - cellular networks provides hardware identity via SIM card; no such hardware assist in traditional Internet
- **confidentiality:** via encryption
- **integrity checks:** digital signatures prevent/detect tampering
- **access restrictions:** password-protected VPNs
- **firewalls:** specialized “middleboxes” in access and core networks:
 - off-by-default: filter incoming packets to restrict senders, receivers, applications
 - detecting/reacting to DOS attacks

... lots more on security (throughout, Chapter 8)

Network under Attack



simplylearn

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Internet history

Introduction: summary

Covered a “ton” of material!

- ❖ Internet overview
- ❖ what’s a protocol?
- ❖ network edge, core, access network
 - packet-switching versus circuit-switching
 - Internet structure
- ❖ performance: loss, delay, throughput
- ❖ layering, service models
- ❖ security
- ❖ history

you now have:

- ❖ context, overview, “feel” of networking
- ❖ more depth, detail *to follow!*

